



Mini Project 02 Report

Fully-Differential Folded Cascode OTA

Analog IC Design | Cadence Virtuoso

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Part 1: g_m/I_D Design Charts

Lab Instruction

Using ADT Sizing Assistant, plot the following design charts vs g_m/I_D for both PMOS and NMOS. Set V_{DS} to a reasonable value.

Answer:

The ADT Sizing Assistant was utilized to generate lookup table plots for the nch_33 and pch_33 2.5V devices.

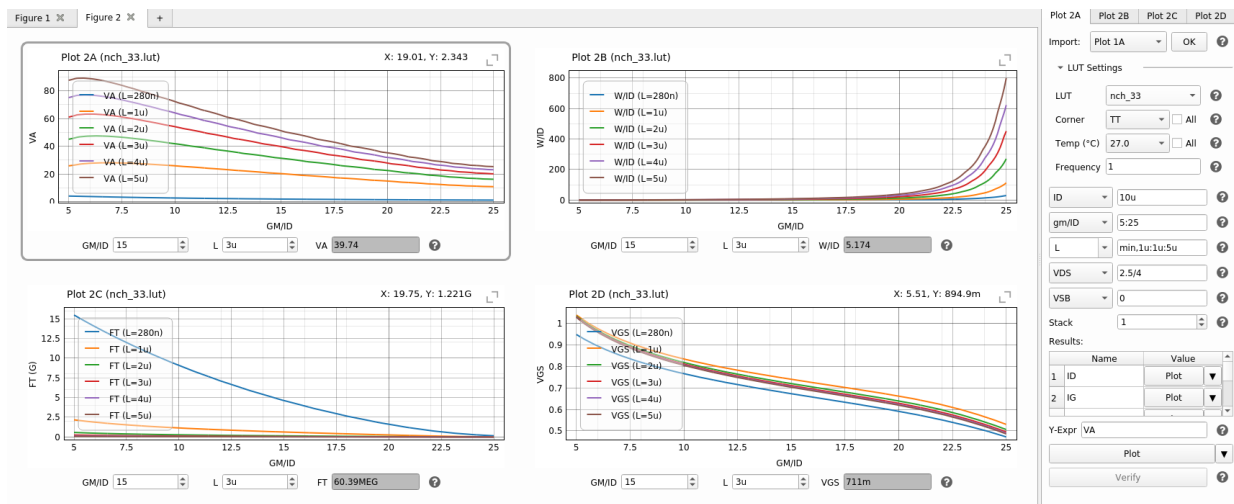


Figure 1. NMOS ADT Charts for V_A , W/I_D , f_T , and V_{GS} .

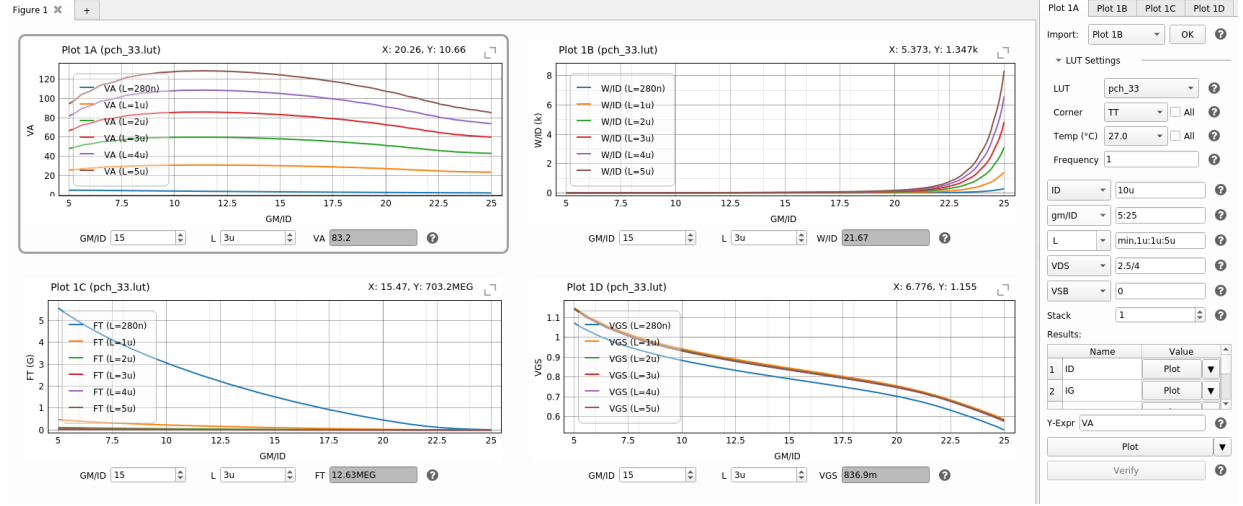


Figure 2. PMOS ADT Charts for V_A , W/I_D , f_T , and V_{GS} .

Part 2: OTA Design Procedure

Lab Instruction

Design a fully-differential folded cascode OTA with capacitive feedback that meets the given specifications. The current of the basic branches should be $10 \mu\text{A}$.

Design Calculations & Hand Analysis

The design procedure adheres to the parameters defined for the GF180MCU 2.5V technology. To optimize for realistic loading, the input pair's parasitic gate capacitance (C_{gg}) is incorporated into the effective load calculation.

1. Settling Time to Bandwidth Conversion:

The requirement dictates a 100 ns settling time for a 1% error margin. Assuming a first-order system response:

$$e^{-t_s/\tau} = 0.01 \implies t_s = -\ln(0.01)\tau \approx 4.6\tau$$

Substituting the required 100 ns settling time to extract the closed-loop bandwidth (ω_{CL}):

$$\tau = \frac{100 \text{ ns}}{4.6} = 21.74 \text{ ns} \implies \omega_{CL} = \frac{1}{\tau} \approx 46 \text{ Mrad/s}$$

2. Feedback Factor (β) & Effective Load with C_{gg} :

To achieve a closed-loop gain magnitude of 2, the feedback network utilizes $C_{in} = 2 \text{ pF}$ and $C_f = 1 \text{ pF}$. The input pair gate capacitance is estimated at $C_{gg} \approx 80 \text{ fF}$, with a $1.5\times$ safety margin (120 fF) applied in parallel with the summing node:

$$C_{in,total} = C_{in} + 1.5C_{gg} = 2 \text{ pF} + 0.12 \text{ pF} = 2.12 \text{ pF}$$

$$\beta = \frac{C_f}{C_{in} + C_f} = \frac{1 \text{ pF}}{2 \text{ pF} + 1 \text{ pF}} = \frac{1}{3}$$

The OTA must drive the explicit 500 fF load in parallel with the series combination of the feedback network:

$$C_{L,eff} = C_L + \left(\frac{C_{in,total} \cdot C_f}{C_{in,total} + C_f} \right) = 500 \text{ fF} + 679.5 \text{ fF} = 1.1795 \text{ pF}$$

3. Required Input Transconductance (g_{m1}):

The Gain-Bandwidth Product (ω_{GBW}) required to sustain the targeted closed-loop dynamics:

$$\omega_{GBW} = \frac{\omega_{CL}}{\beta} = 46 \text{ Mrad/s} \times 3 = 138 \text{ Mrad/s}$$

$$g_{m1} = \omega_{GBW} \times C_{L,eff} = 138\text{M} \times 1.1795\text{p} \approx 162.8 \mu\text{S}$$

4. Branch Currents (I_D) & g_m/I_D Target:

With the fundamental branch current constrained to 10 μA , the necessary g_m/I_D target is established:

$$g_m/I_D = \frac{162.8 \mu\text{S}}{10 \mu\text{A}} \approx 16.3 \text{ V}^{-1}$$

- **Input Branches:** 10 μA each
- **Cascode Branches:** 10 μA each
- **Top Tail Source:** Must supply both input branches (20 μA total)
- **Bottom Sinks:** Must sink both the input and cascode branches (20 μA each)

Transistor Sizing & Width Extraction

To exchange excess open-loop gain for reduced layout area, the cascode channel lengths were optimized from 420 nm down to 350 nm. Device widths were subsequently extracted using ADT based on the refined 16.3 V^{-1} operational target.

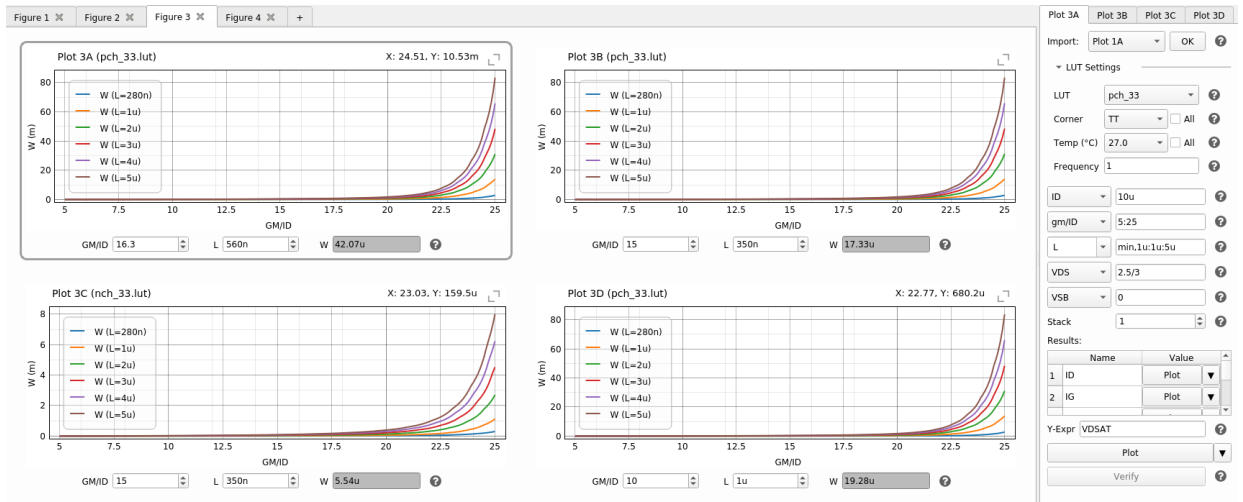


Figure 3. ADT width extraction for PMOS Input Pair ($L = 560 \text{ nm}$), PMOS Cascodes ($L = 350 \text{ nm}$), NMOS Cascodes ($L = 350 \text{ nm}$), and PMOS Sources.

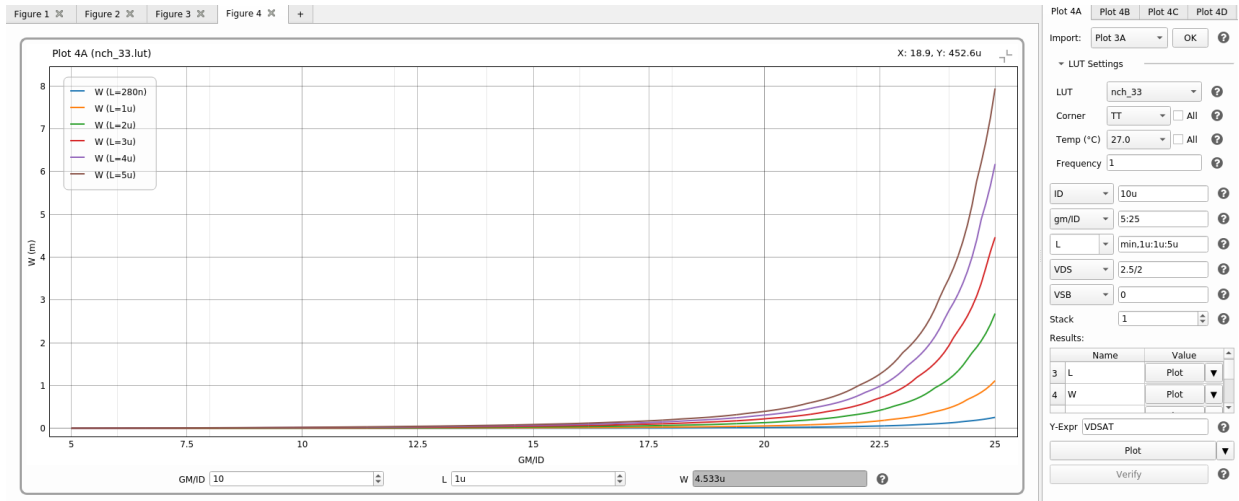


Figure 4. ADT width extraction for the NMOS Current Sources ($10 \mu\text{A}$ unit $W = 4.53 \mu\text{m}$).

Circuit Role	Instances	Length (L)	Width (W)	Multiplier (m)
PMOS Input Pair	M7, M8	560 nm	$42.07 \mu\text{m}$	1
PMOS Top Cascodes	M5, M6	350 nm	$17.33 \mu\text{m}$	1
NMOS Bottom Cascodes	M9, M11	350 nm	$5.54 \mu\text{m}$	1
PMOS Top Sources ($10 \mu\text{A}$)	M2, M4	$1 \mu\text{m}$	$19.28 \mu\text{m}$	1
PMOS Tail Bias ($20 \mu\text{A}$)	M3	$1 \mu\text{m}$	$19.28 \mu\text{m}$	2
NMOS Bottom Sinks ($20 \mu\text{A}$)	M12, M15	$1 \mu\text{m}$	$4.53 \mu\text{m}$	2

Table 1. Final OTA Device Geometries.

Cadence Schematic Implementation

The finalized geometries were implemented in the core Cadence Virtuoso schematic below.

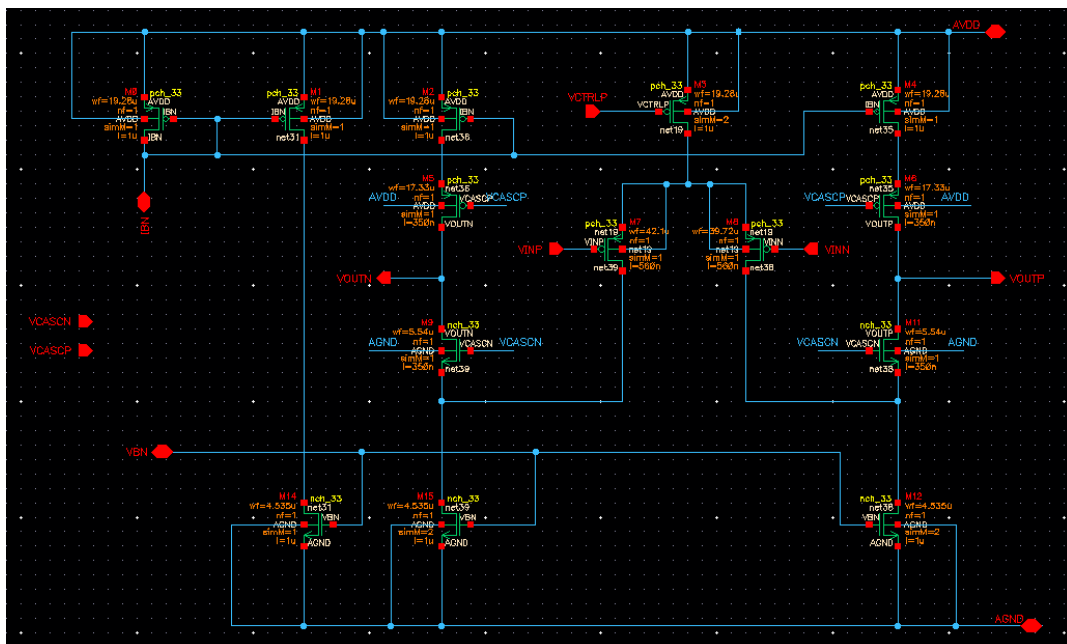


Figure 5. Fully-Differential Folded Cascode OTA Schematic with parameterized device sizing.

Cascode Biasing (V_{CASCP} and V_{CASCN})

Lab Instruction

From the assumed g_m/I_D , select suitable biasing for the cascode transistors (V_{CASCP} and V_{CASCN}). Hint: Set $V_{CASCN} \approx V_{GS,casc} + V_{current_source}^*$.

Answer:

1. Initial Theoretical Calculations

To keep the current mirrors deep in saturation while maximizing output swing, the theoretical initial cascode bias voltages were extracted directly from the ADT Sizing Assistant logs for the optimized $L = 350$ nm cascodes and $L = 1\mu\text{m}$ current sources.

PMOS Extraction Data:

Info: [Sizing Assistant] The resultant point 'VSG = 826.9m'
Info: [Sizing Assistant] The resultant point 'VSTAR = 202.2m'

NMOS Extraction Data:

Info: [Sizing Assistant] The resultant point 'VGS = 717.8m'
Info: [Sizing Assistant] The resultant point 'VSTAR = 202.6m'

The theoretical NMOS cascode bias is calculated using the V_{GS} of the NMOS cascodes (M9, M11) and the V^* of the NMOS bottom sinks (M15, M12):

$$V_{CASCN} = V_{GS_M9,11} + V_{M15,12}^*$$

$$V_{CASCN} = 717.8 \text{ mV} + 202.6 \text{ mV} = \mathbf{0.9204 \text{ V}}$$

The theoretical PMOS cascode bias is calculated from the 2.5 V supply, subtracting the V_{SG} of the PMOS cascodes (M5, M6) and the V^* of the top PMOS sources (M2, M4):

$$V_{CASCP} = V_{DD} - (V_{SG_M5,6} + V_{M2,4}^*)$$

$$V_{CASCP} = 2.5 \text{ V} - (0.8269 \text{ V} + 0.2022 \text{ V}) = \mathbf{1.4709 \text{ V}}$$

2. Simulation-Based Bias Tuning

While the theoretical calculations provided an excellent baseline, initial simulations revealed that these exact values placed the current mirror sources (M12, M15, M4) on the absolute boundary of the triode region ($V_{DS} \approx V_{DSAT}$).

To ensure saturation under process variations, explicit functional safety margins were introduced to the final testbench variables:

- V_{CASCN} was increased to **1.1 V**. This pulls the bottom folding node higher, granting the NMOS sinks roughly 140 mV of V_{DS} headroom above V_{DSAT} .
- V_{CASCP} was decreased to **1.3 V**. This pushes the top folding node lower, supplying the PMOS sources with roughly 120 mV of adequate compliance voltage.

3. Output Swing Verification (KVL Analysis)

Adding cascode margin inherently degrades the maximum theoretical output swing. However, KVL analysis confirms this trade-off is safe for the specified limits. The required differential output swing is 1.2 V_{pk-pk} , equating to a single-ended excursion of $\pm 0.3 \text{ V}$ around the 1.25 V common-mode target. Thus, the required absolute output range is $[0.95 \text{ V}, 1.55 \text{ V}]$.

The lower swing limit ($V_{out,min}$) is bounded by the NMOS cascode (M9) entering triode:

$$V_{out,min} \approx V_{CASCN} - V_{GS,M9} + V_{DSAT,M9}$$

Since $V_{GS} - V_{DSAT} \approx V_{TH}$, this simplifies to:

$$V_{out,min} \approx V_{CASCN} - V_{TH,M9} = 1.1 \text{ V} - 0.757 \text{ V} = \mathbf{0.343 \text{ V}}$$

With a requirement of 0.95 V , the design maintains over 600 mV of excess margin on the lower rail.

The upper swing limit ($V_{out,max}$) is bounded by the PMOS cascode (M5) entering triode:

$$V_{out,max} \approx V_{CASCN} + |V_{GS,M5}| - |V_{DSAT,M5}| \approx V_{CASCN} + |V_{TH,M5}|$$

$$V_{out,max} \approx 1.3 \text{ V} + 0.866 \text{ V} = \mathbf{2.166 \text{ V}}$$

With a requirement of 1.55 V , the design preserves over 600 mV of excess margin on the upper rail. The tuned biases guarantee deep saturation for all current mirrors without violating the output swing specifications.

Part 3: Open-Loop OTA Simulation (Behavioral CMFB)

Lab Instruction

Report the following: 1) Schematic of the OTA and bias circuit with DC node voltages and transistors OP parameters (id, vgs, vds, vdsat, vth, gm, gds, region) clearly annotated.

- Set V_{ICM} at the middle of the CMIR.
- Select V_{REF} to maximize the symmetrical output swing.
- What is the CM level at the OTA output?
- What are the differential input and output voltages of the error amplifier? What is the relation between them?

Testbench Configuration & The Replica Biasing Technique

The behavioral CMFB utilizes an ideal VCVS error amplifier with a gain of 1 to avoid loading the OTA output. However, using a unity-gain loop creates a massive steady-state error if the amplifier's negative terminal is simply tied to ground.

Here is the core problem: the standard VCVS equation is $V_{CTRLP} = 1 \times (V_{CM} - V_{REF})$. For the OTA to function properly, the top PMOS current sources need a specific DC gate voltage (around 1.557 V) to supply the correct tail current. If the error amplifier is referenced to ground, the only way it can output 1.557 V is if the error term $(V_{CM} - V_{REF})$ itself equals 1.557 V. This would force the actual output common-mode (V_{CM}) to shift wildly away from our 1.25 V target, instantly pushing the amplifier into the supply rails.

To fix this without artificially increasing the behavioral amplifier's gain, we use a replica biasing technique as discussed in the lectures. Professor Hesham explained that the concept relies on introducing a "replica" of the PMOS tail transistor to generate the exact quiescent bias voltage (V_{BP}) required. Instead of referencing the error amplifier to ground (0 V), we reference it to this replica voltage.

In our testbench, we implemented this seamlessly by tapping the gate voltage directly from the primary PMOS bias diode (the I_{BN} node) in our main bias network. This node naturally provides the exact nominal voltage (1.557 V) needed by the PMOS tail. We wired this node directly to the negative terminal of the ideal error amplifier.

With this setup, the closed-loop equation becomes:

$$V_{CTRLP} = V_{BP} + 1 \times (V_{CM} - V_{REF})$$

Because the amplifier now pivots around the correct baseline voltage, when the output common-mode matches the reference ($V_{CM} = V_{REF}$), the mathematical error drops to zero. The amplifier naturally outputs the exact replica voltage ($V_{CTRLP} = V_{BP}$), keeping the OTA perfectly biased while locking the output common-mode exactly to 1.25 V with zero steady-state error.

Additionally, to provide a valid DC path to ground and satisfy the simulator's matrix solver requirements for the ideal baluns, 1 T Ω dummy resistors were attached to the V_{OCM} and V_{ODIFF} nets.

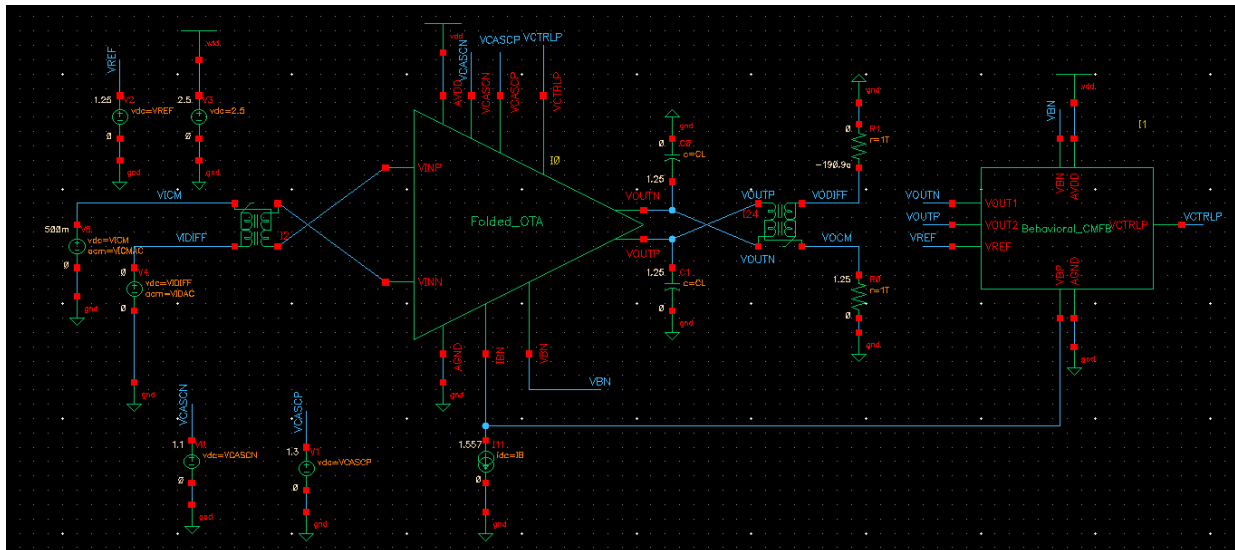


Figure 6. Behavioral CMFB Testbench incorporating replica biasing for zero steady-state error.

Transistor Operating Parameters

The DC operating parameters were tabulated following simulation. The applied safety margins to the cascode gate voltages ensure that all active devices operate robustly within the saturation region (Region 2).

Device	Role	I_D (μA)	V_{GS} (mV)	V_{DS} (V)	V_{DSAT} (mV)	V_{TH} (mV)	g_m (μS)	g_{ds} (μS)	Reg
M7, M8	PMOS Input Pair	-10.05	-831.4	-1.021	-92.3	-789.9	163.9	0.49	2
M5, M6	PMOS Cascodes	-9.75	-922.9	-0.973	-112.3	-866.3	148.4	1.57	2
M9, M11	NMOS Cascodes	9.75	789.2	0.939	103.9	757.2	146.2	2.54	2
M12, M15	NMOS Bottom Sinks	19.79	835.9	0.311	172.6	687.3	196.5	1.76	2
M3	PMOS Tail Bias	-20.10	-943.0	-1.169	-155.4	-783.9	201.8	0.39	2
M1	PMOS Source (Left)	-10.13	-942.9	-1.664	-155.4	-783.9	101.7	0.16	2
M4	PMOS Source (Right)	-9.74	-942.9	-0.277	-155.4	-783.9	97.4	1.12	2
M0	PMOS Bias Diode	-10.00	-942.9	-0.943	-155.4	-783.9	100.4	0.23	2
M2	PMOS Bias to NMOS	-9.74	-942.9	-0.277	-155.4	-783.9	97.4	1.12	2
M14	NMOS Bias Diode	10.13	835.9	0.836	172.6	687.3	100.6	0.33	2

Table 2. DC Operating Parameters demonstrating all transistors are actively biased in saturation (Region 2).

DC Operating Point Results

- **Input Settings:** V_{ICM} was anchored at 500 mV, centering it squarely within the CMIR specifications. V_{REF} was designated as 1.25 V to maximize the theoretical symmetrical swing output across the 2.5 V supply.
- **Output CM Level:** Based on the schematic annotations, the output common-mode voltage (V_{OUTN} and V_{OUTP}) locked precisely to the 1.25 V target. The lack of observable offset verifies the mathematical efficacy of the replica-biasing technique.
- **Error Amplifier Voltages & Relations:**
 - **Inputs:** The error amplifier processes two inputs: the sensed actual common-mode voltage (V_{CM}) and the objective reference voltage (V_{REF}). The functional **differential input** is mathematically defined as $(V_{CM} - V_{REF})$.
 - **Output:** The resulting output is the single-ended control voltage (V_{CTRLP}) utilized to govern the PMOS tail array.
 - **Relation:** By referencing the negative terminal to the replica bias (V_{BP}), the block structurally adds the baseline voltage to the computed error. Given $A_v = 1$, the definitive operational relationship is:

$$V_{CTRLP} = V_{BP} + (V_{CM} - V_{REF})$$

From the simulation state: $1.557 \text{ V} = 1.557 \text{ V} + (1.25 \text{ V} - 1.25 \text{ V})$.

Differential Small-Signal Characteristics

To evaluate the open-loop differential performance, an AC analysis (1Hz to 10GHz) was executed. The differential input source (V_{IDIFF}) was configured with an AC magnitude of 1 V, and the common-mode AC magnitude (V_{ICMAC}) was set to 0 V to strictly isolate the differential transfer function. The resulting magnitude and phase Bode plots for the output differential net (V_{ODIFF}) are presented below.

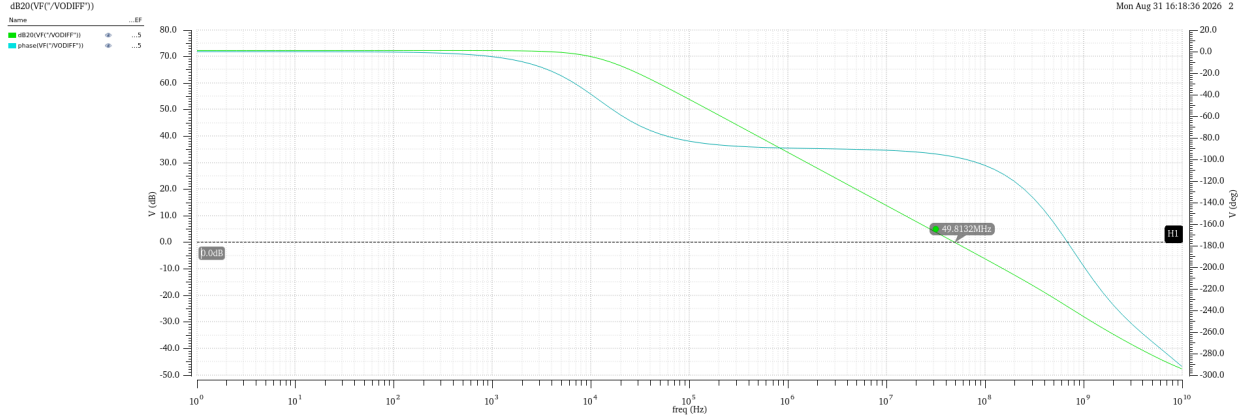


Figure 7. Open-loop differential AC magnitude (dB) and phase response.

Test	Output	Nominal	Spec	Weight	Pass/Fail
AC_Analysis	dB20(VF("/VODIFF"))				
AC_Analysis	phase(VF("/VODIFF"))				
AC_Analysis	DC Gain	4.1k			
AC_Analysis	Ao in dB	72.25			
AC_Analysis	BW	12.15k			
AC_Analysis	UGF	49.85M			
AC_Analysis	PM	82.32			

Figure 8. Cadence calculator evaluations for the differential AC performance metrics (Behavioral CMFB).

Theoretical Hand Calculations

To mathematically validate the simulated results, the theoretical small-signal parameters are derived using the operating point data extracted in the previous section.

1. Total Open-Loop DC Gain (A_o):

The intrinsic resistance looking into the top and bottom folding nodes establishes the degeneration resistance for the cascode devices.

$$R_{S,cascP} = r_{o,M7} || r_{o,M4} = \left(\frac{1}{0.49 \mu S} \right) || \left(\frac{1}{1.12 \mu S} \right) = 620.9 \text{ k}\Omega$$

$$R_{S,cascN} = r_{o,M15} = \frac{1}{1.76 \mu S} = 568.1 \text{ k}\Omega$$

The absolute output impedance looking into the cascode drains incorporates their intrinsic gain:

$$R_{up} \approx (g_{m,M6} + g_{mb,M6})r_{o,M6}R_{S,cascP} = (197.25 \mu S) \left(\frac{1}{1.57 \mu S} \right) (620.9 \text{ k}\Omega) = 78.0 \text{ M}\Omega$$

$$R_{down} \approx (g_{m,M11} + g_{mb,M11})r_{o,M11}R_{S,cascN} = (180.81 \mu S) \left(\frac{1}{2.54 \mu S} \right) (568.1 \text{ k}\Omega) = 40.4 \text{ M}\Omega$$

The differential DC gain evaluates to:

$$R_{out} = R_{up} || R_{down} = 78.0 \text{ M}\Omega || 40.4 \text{ M}\Omega = \mathbf{26.6 \text{ M}\Omega}$$

$$A_{o(\text{linear})} = g_{m1} \times R_{out} = 163.9 \text{ }\mu\text{S} \times 26.6 \text{ M}\Omega = \mathbf{4360 \text{ V/V}} \implies \mathbf{72.8 \text{ dB}}$$

2. Dominant Pole (p_1), BW, & GBW:

With an explicit differential load capacitance of $C_L = 500 \text{ fF}$ per side, the theoretical bandwidth dynamics evaluate as follows:

$$BW = \frac{1}{2\pi R_{out} C_L} = \frac{1}{2\pi (26.6 \text{ M}\Omega)(500 \text{ fF})} = \mathbf{11.9 \text{ kHz}}$$

$$GBW = A_{o(\text{linear})} \times BW = 4360 \times 11.9 \text{ kHz} = \mathbf{51.9 \text{ MHz}}$$

3. Non-Dominant Poles (p_2, p_3) & Phase Margin (PM):

To calculate Phase Margin, the non-dominant poles at the NMOS and PMOS folding nodes must be extracted by summing the parasitic capacitances ($C_{db}, C_{gd}, C_{gs}, C_{sb}$) from the OP data.

$$C_{fold,N} \approx C_{db,M7} + C_{gd,M7} + C_{db,M15} + C_{gd,M15} + C_{gs,M9} + C_{sb,M9} \approx \mathbf{26.8 \text{ fF}}$$

$$f_{p2} \approx \frac{g_{m,M9} + g_{mb,M9}}{2\pi C_{fold,N}} = \frac{180.8 \text{ }\mu\text{S}}{2\pi (26.8 \text{ fF})} = \mathbf{1.07 \text{ GHz}}$$

$$C_{fold,P} \approx C_{db,M1} + C_{gd,M1} + C_{gs,M5} + C_{sb,M5} \approx \mathbf{32.1 \text{ fF}}$$

$$f_{p3} \approx \frac{g_{m,M5} + g_{mb,M5}}{2\pi C_{fold,P}} = \frac{197.2 \text{ }\mu\text{S}}{2\pi (32.1 \text{ fF})} = \mathbf{980 \text{ MHz}}$$

The theoretical Phase Margin degrades from 90° according to the proximity of these non-dominant poles to the GBW:

$$PM \approx 90^\circ - \arctan\left(\frac{GBW}{f_{p2}}\right) - \arctan\left(\frac{GBW}{f_{p3}}\right)$$

$$PM \approx 90^\circ - \arctan\left(\frac{51.9}{1070}\right) - \arctan\left(\frac{51.9}{980}\right) = 90^\circ - 2.77^\circ - 3.03^\circ = \mathbf{84.2^\circ}$$

Simulation vs. Theoretical Comparison

The exact performance metrics were quantified using Cadence calculator expressions evaluating the AC sweep buffer data. The empirical simulation values exhibit an exceptional correlation with the comprehensive hand calculations, verifying the robustness of the extracted parasitic models.

Parameter	Hand Calculation	Simulation	Deviation
DC Gain (A_o , linear)	4360 V/V	4.1k V/V	6.3%
DC Gain (A_o , dB)	72.8 dB	72.25 dB	0.7%
Bandwidth (BW)	11.9 kHz	12.15 kHz	2.1%
Gain-Bandwidth (GBW)	51.9 MHz	49.8 MHz	4.2%
Phase Margin (PM)	84.2°	82.32°	2.2%

Table 3. Comparison matrix of hand-calculated vs. simulated open-loop AC characteristics.

Part 4: Transistor-Level CMFB Design

Lab Instruction

Replace the ideal behavioral CMFB block with a physical transistor-level circuit. Design the PMOS source followers, the sensing resistor network, and the error amplifier. Justify your device sizing, current budgeting, and demonstrate the DC operating point stability.

Design Strategy & Output Swing Constraints

When transitioning from the ideal behavioral model to a physical circuit, two fundamental limitations dictate the CMFB topology: the level-shifting swing headroom and tail current stability.

1. Output Common-Mode Reference (V_{REF}) Re-Evaluation:

In the behavioral testbench, V_{REF} was centered at 1.25 V. However, the PMOS Common-Drain (CD) buffers introduce an intrinsic upward level shift of $|V_{GSP}| \approx 845$ mV. To keep the PMOS current sources (M_0, M_5, M_{10}) above the buffers in saturation, the maximum output voltage is bounded by:

$$V_{out,max} = V_{DD} - V_{CS}^* - |V_{GS,CD}|$$

Assuming $V_{CS}^* \approx 200$ mV and $|V_{GS,CD}| \approx 850$ mV at 2.5 V supply:

$$V_{out,max} \approx 2.5 \text{ V} - 0.2 \text{ V} - 0.85 \text{ V} = \mathbf{1.45 \text{ V}}$$

The lower swing limit is bounded by the NMOS cascode and bottom sink entering triode:

$$V_{out,min} \approx V_{DSAT,casc} + V_{DSAT,sink} \approx 0.11 \text{ V} + 0.17 \text{ V} \approx \mathbf{0.28 \text{ V} - 0.35 \text{ V}}$$

To center the output swing symmetrically within this practical range [0.35 V, 1.45 V], the output common-mode target was re-calculated:

$$V_{OCM,center} \approx \frac{1.45 \text{ V} + 0.35 \text{ V}}{2} = 0.90 \text{ V} \implies \mathbf{V_{REF} = 0.95 \text{ V}}$$

Anchoring V_{REF} at 0.95 V accommodates the required differential swing of 1.2 V_{pk-pk} ($\pm 0.3 \text{ V}$ single-ended, spanning 0.65 V to 1.25 V) while maintaining over 200 mV of headroom from both saturation boundaries.

2. 80% / 20% Tail Biasing Split:

Allocating 100% of the OTA tail current to the CMFB error amplifier poses severe operational risks: startup transients or error amplifier saturation can pull $V_{CTRLP} \rightarrow V_{DD}$, starving the input differential pair and causing latch-up. To guarantee a robust quiescent baseline, the main PMOS tail source was partitioned using an 80/20 ratio:

- **Fixed Quiescent Source (80%, 16 μA):** Sized with $W = 7.71 \mu\text{m}$ and multiplier $m = 4$. Its gate is tied directly to the constant bias line (IBN / V_{BP}), ensuring the input pair can never starve.
- **Dynamic CMFB Trimming Source (20%, 4 μA):** Sized with $W = 7.71 \mu\text{m}$ and multiplier $m = 1$. Its gate is driven by V_{CTRLP} , dedicating 4 μA of dynamic range for common-mode regulation.

3. CMFB Branch Current Budgeting (20 μA Total):

The overall CMFB consumption was set to 20 μA (50% of the main OTA's core branch current):

- **Sensing Buffers (Left & Right):** 4 μA each.
- **Dummy Reference Buffer:** 4 μA (to identically track process and temperature shifts on V_{REF}).
- **Error Amplifier Differential Pair:** 8 μA total (4 μA per branch at equilibrium).

With 4 μA per buffer branch, the sensing resistors were set to $R = 100\text{ k}\Omega$. The peak dynamic leakage current during full differential excursion is $I_{leak} = 0.6\text{ V}/(2 \times 100\text{ k}\Omega) = 3\text{ }\mu\text{A}$, leaving a safe 1 μA margin that completely prevents buffer starvation.

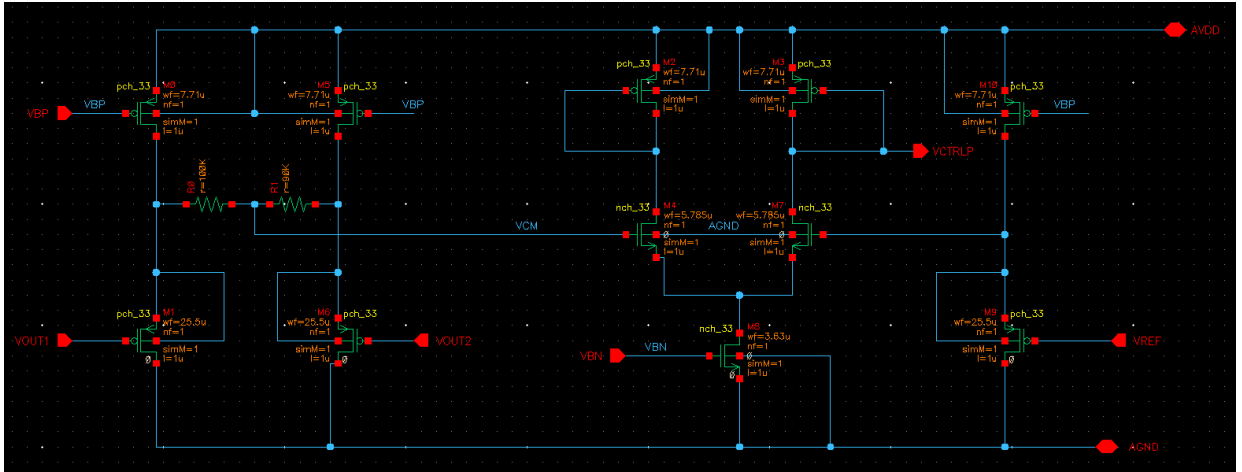


Figure 9. Transistor-level CMFB schematic with DC operating voltages annotated ($V_{REF} = 0.95\text{ V}$).

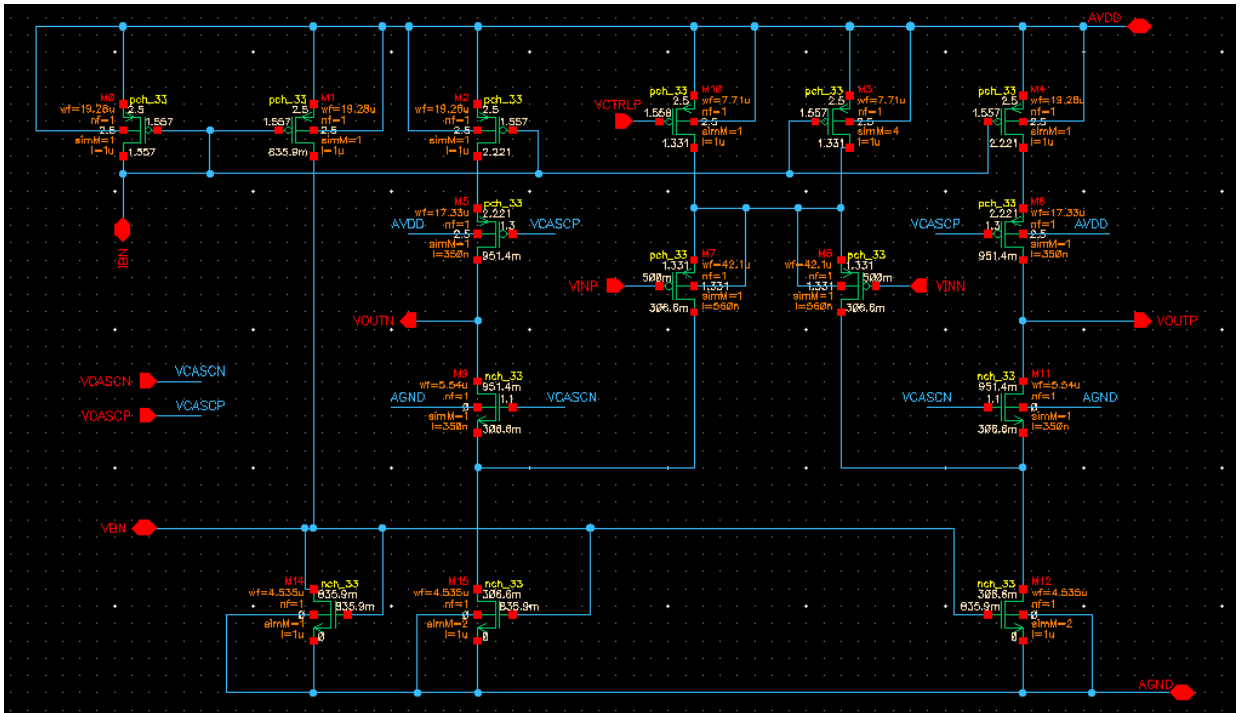


Figure 10. Main Folded Cascode OTA schematic incorporating the 80/20 tail split (M3 with $m = 4$, M10 with $m = 1$).

DC Operating Point Analysis

The DC operating point parameters for both the revised Folded OTA and the transistor-level CMFB circuit were extracted to confirm active operation in saturation (Region 2) across all devices.

Device	Role	I_D (μA)	$ V_{GS} $ (mV)	$ V_{DS} $ (V)	$ V_{DSAT} $ (mV)	$ V_{TH} $ (mV)	g_m (μS)	g_{ds} (μS)	Reg
Main Folded OTA (80/20 Tail Architecture)									
M7, M8	PMOS Input Pair	10.04	831.3	1.025	92.2	789.9	163.8	0.49	2
M5, M6	PMOS Cascodes	9.75	920.6	1.269	111.5	865.3	149.0	1.47	2
M9, M11	NMOS Cascodes	9.74	793.4	0.645	105.2	759.1	145.9	2.71	2
M12, M15	NMOS Sinks ($m = 2$)	19.78	835.9	0.307	172.6	687.3	196.4	1.82	2
M3	PMOS Fixed Tail ($m = 4$)	16.08	942.9	1.169	155.5	783.8	161.2	0.32	2
M10	PMOS CMFB Tail ($m = 1$)	4.00	942.5	1.169	155.2	783.8	40.17	0.08	2
M1	PMOS Source (Left)	10.13	942.9	1.664	155.4	783.9	101.7	0.16	2
M2, M4	PMOS Source (Right)	9.74	942.9	0.279	155.4	783.9	97.4	1.09	2
M0	PMOS Bias Diode	10.00	942.9	0.943	155.4	783.9	100.4	0.23	2
M14	NMOS Bias Diode	10.13	835.9	0.836	172.6	687.3	100.6	0.33	2
Transistor-Level CMFB									
M1, M6, M9	PMOS CD Buffers	3.98	845.1	1.797	94.2	783.9	60.33	0.07	2
M0, M5, M10	PMOS Bias Sources	3.98	942.9	0.704	155.5	783.8	39.87	0.12	2
M4, M7	NMOS Inputs	4.02	998.3	0.759	114.3	944.4	60.37	0.16	2
M2, M3	PMOS Diode Loads	4.02	943.4	0.943	155.8	783.8	40.22	0.09	2
M8	NMOS Tail Sink	8.04	835.9	0.798	172.4	686.8	79.69	0.27	2

Table 4. DC Operating Parameters demonstrating proper saturation (Region 2) across both the OTA and the physical CMFB loop under the 80/20 architecture.

Lab Report Questions & Justifications

1. What is the CM level at the OTA output? Why?

The common-mode level at the differential outputs (V_{OUTN} and V_{OUTP}) locked precisely to **951.4 mV**. This tracks the $V_{REF} = 950$ mV objective with a microscopic steady-state error of only **+1.4 mV**. The loop actively senses the physical output common mode through the PMOS buffers and drives the dynamic 20% tail device (M_{10} in the OTA) via V_{CTRLP} to balance the current sinks. The 1.4 mV offset is the natural consequence of finite DC loop gain in the error amplifier and channel-length modulation.

2. What are the differential input and output voltages of the error amplifier? What is the relation between them?

- **Differential Input:** The inputs to the NMOS error amplifier are the level-shifted common-mode voltage ($V_{CM,shift} = 1.797$ V at the gate of M4) and the identically level-shifted reference ($V_{REF,shift} = 1.795$ V at the gate of M7). The resulting differential error input is:

$$V_{diff,in} = V_{CM,shift} - V_{REF,shift} = 1.797 \text{ V} - 1.795 \text{ V} = \mathbf{+2.0 \text{ mV}}$$

- **Output Voltage:** The single-ended control voltage (V_{CTRLP}) generated at the drain of M7 settled at **1.558 V**, perfectly matching the nominal PMOS gate bias ($V_{BP} = 1.557$ V).

- **Transfer Relationship:** The error amplifier governs the dynamic control node according to:

$$V_{CTRLP} = V_{CTRLP,Q} + A_{v,err} \times (V_{CM,shift} - V_{REF,shift})$$

Because diode-connected PMOS loads (M2, M3) are utilized, the small-signal voltage gain is determined by the transconductance ratio:

$$A_{v,err} \approx \frac{1}{2} \left(\frac{g_{m,M7}}{g_{m,M3}} \right) \approx \frac{1}{2} \left(\frac{60.37 \mu\text{S}}{40.22 \mu\text{S}} \right) \approx \mathbf{0.75 \text{ V/V}}$$

The amplifier applies this fractional gain directly to the +2 mV input discrepancy, establishing zero control authority clipping.

Differential Small-Signal Characteristics (Transistor-Level CMFB)

To evaluate the differential AC performance with the transistor-level CMFB and 80/20 split active, an AC analysis was executed from 1 Hz to 10 GHz ($V_{IDAC} = 1 \text{ V}$, $V_{ICMAC} = 0 \text{ V}$).

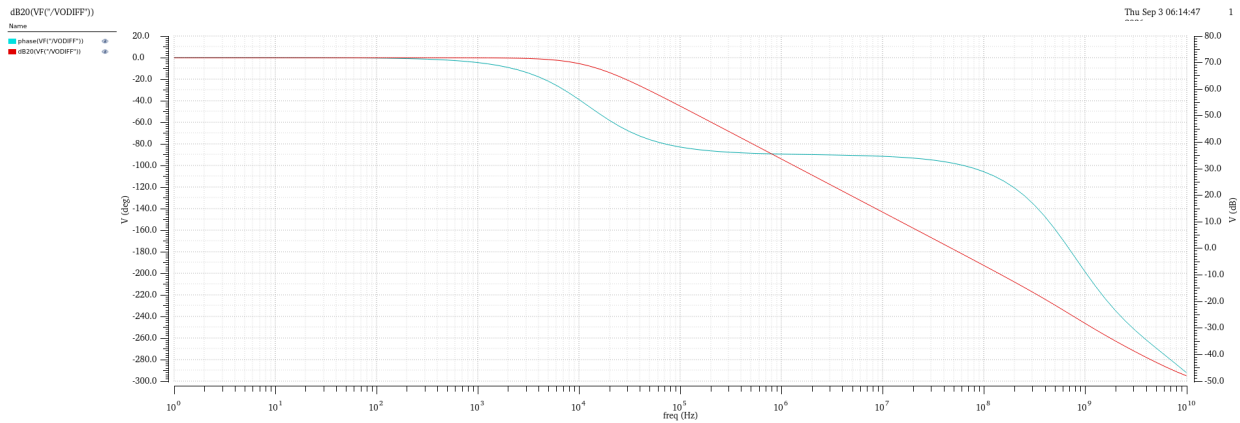


Figure 11. Open-loop differential AC magnitude and phase response with the 80/20 transistor-level CMFB active.

Test	Output	Nominal	Spec	Weight	Pass/Fail
AC_Analysis	Ao Daingain	3.894k			
AC_Analysis	Ao in dB	71.81			
AC_Analysis	BW	12.39k			
AC_Analysis	UGF	48.54M			
AC_Analysis	GBW	48.36M			
AC_Analysis	PM	82.32			
AC_Analysis	dB20(VF("VODIFF"))				
AC_Analysis	phase(VF("VODIFF"))				

Figure 12. Cadence calculator evaluations for open-loop differential AC metrics.

Parameter	Behavioral CMFB (Part 3)	Physical CMFB (Part 4)	Deviation
DC Gain (A_o , linear)	4100 V/V	3894 V/V	-5.0%
DC Gain (A_o , dB)	72.25 dB	71.81 dB	-0.6%
Bandwidth (BW)	12.15 kHz	12.39 kHz	+1.9%
Gain-Bandwidth Product (GBW)	49.80 MHz	48.36 MHz	-2.8%
Unity Gain Frequency (UGF)	49.80 MHz	48.54 MHz	-2.5%
Phase Margin (PM)	82.32°	82.32°	0.0%

Table 5. Comparison matrix of differential AC performance: Behavioral vs. Physical 80/20 CMFB.

Part 5: Closed-Loop Simulation (AC and STB Analysis)

Lab Instruction

Evaluate the fully-differential folded cascode OTA in a closed-loop inverting capacitive feedback configuration. Verify closed-loop differential AC metrics ($A_{cl} \geq 2$), assess the DC operating point under feedback, and execute STB stability analyses on both the differential and CMFB feedback loops.

Closed-Loop DC Operating Point

The OTA was embedded in a closed-loop capacitive feedback topology utilizing nominal capacitances of $C_{in} = 2$ pF and $C_f = 1$ pF to target an ideal closed-loop voltage gain of $A_{cl} = C_{in}/C_f = 2$ V/V (6.02 dB). Continuous DC bias paths to the high-impedance gates were established via ultra-high feedback resistors ($R_f \approx 1$ T Ω) placed in parallel with C_f .

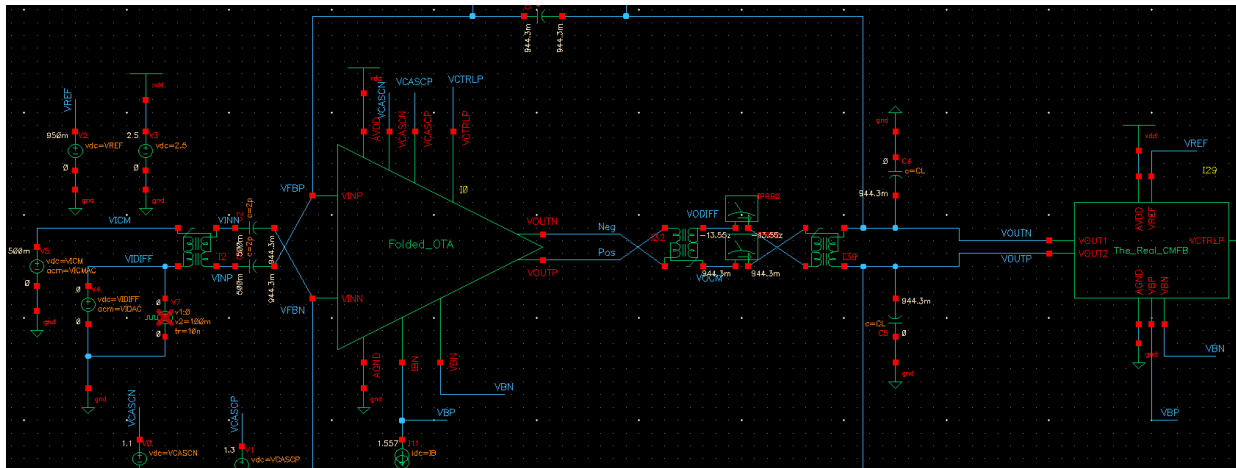


Figure 13. Closed-loop testbench schematic annotated with steady-state DC operating voltages ($V_{REF} = 0.95$ V).

Lab Report DC Questions & Justifications

1. What is the CM level at the OTA output? Why?

The steady-state common-mode voltage at the differential outputs (V_{OUTN} and V_{OUTP}) locked precisely to **944.3 mV**. The active CMFB network continuously senses this common-mode voltage through the PMOS source-follower buffers and drives the dynamic tail trim device (M_{10}) via V_{CTRLP} to track the $V_{REF} = 0.95$ V reference. The microscopic 5.7 mV discrepancy from nominal V_{REF} is a natural consequence of the finite DC open-loop gain of the CMFB error amplifier combined with minute sub-threshold gate leakage currents across the 1 T Ω feedback resistors.

2. What is the CM level at the OTA input? Why?

The common-mode voltage at the OTA input gates (V_{INP} and V_{INN}) settled to exactly **944.3 mV**, matching the output common-mode level identically. The input common-mode source ($V_{ICM} = 500$ mV) applied ahead of the input balun is purely AC-coupled and completely blocked from the amplifier by the series input capacitors ($C_{in} = 2$ pF). Consequently, the DC gate operating point is established entirely through the 1 T Ω feedback resistors connecting the outputs directly back to the inputs. Because MOSFET gates draw negligible DC current ($I_G \approx 0$), the voltage drop across the feedback resistors is zero ($I_G R_f \approx 0$), forcing the input common-mode level to float directly to the output common-mode potential ($V_{IN,cm} = V_{OUT,cm} = 944.3$ mV).

Differential Closed-Loop AC Response

A small-signal differential AC sweep was executed from 1 Hz to 10 GHz ($V_{IDAC} = 1$ V, $V_{ICMAC} = 0$ V).

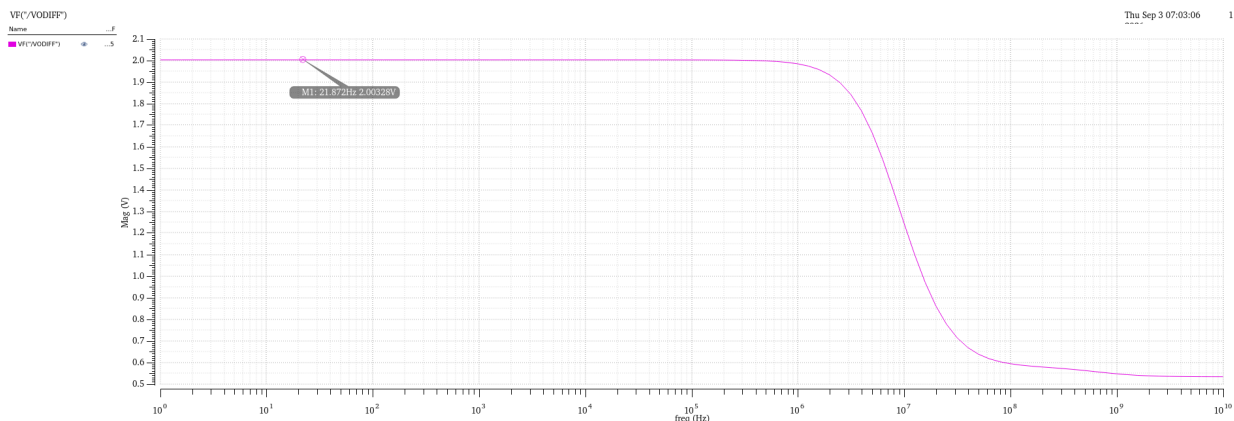


Figure 14. Closed-loop differential AC transfer function magnitude (V_{ODIFF}) versus frequency.

Test	Output	Nominal	Spec	Weight	Pass/Fail
AC_Analysis	VF("VODIFF")				
AC_Analysis	dB20(VF("VODIFF"))				
AC_Analysis	Acl	2.003			
AC_Analysis	Acl dB	6.035			
AC_Analysis	BW	7.708M			
AC_Analysis	UGF	15.12M			
AC_Analysis	PM	83.3			
AC_Analysis	GBW Closed loop	15.44M			

Figure 15. Cadence calculator measurements for the closed-loop differential AC response.

Metric	Design Specification	Simulation Result	Compliance
Closed-Loop DC Gain (A_{cl})	≥ 2.0 V/V (6.02 dB)	2.003 V/V (6.035 dB)	Exceeded (Pass)
Closed-Loop Bandwidth (BW_{CL})	—	7.708 MHz	—
Unity Gain Frequency (UGF)	—	15.12 MHz	—
Closed-Loop GBW ($A_{cl} \times BW_{CL}$)	—	15.44 MHz	Robust

Table 6. Differential closed-loop AC characteristics confirming compliance with the $A_{cl} \geq 2$ target.

The closed-loop DC gain of **2.003** V/V fully satisfies the primary project specification ($A_{cl} \geq 2$). The feedback loop maintains flat differential gain across the baseband spectrum, exhibiting a -3 dB closed-loop bandwidth of 7.708 MHz.

Differential and CMFB Loops Stability (STB Analysis)

To extract loop gain and phase margin without disturbing closed-loop DC operating points, an STB analysis was executed using isolated current probes (iprobe) inserted into the differential and common-mode signal paths between the output baluns.

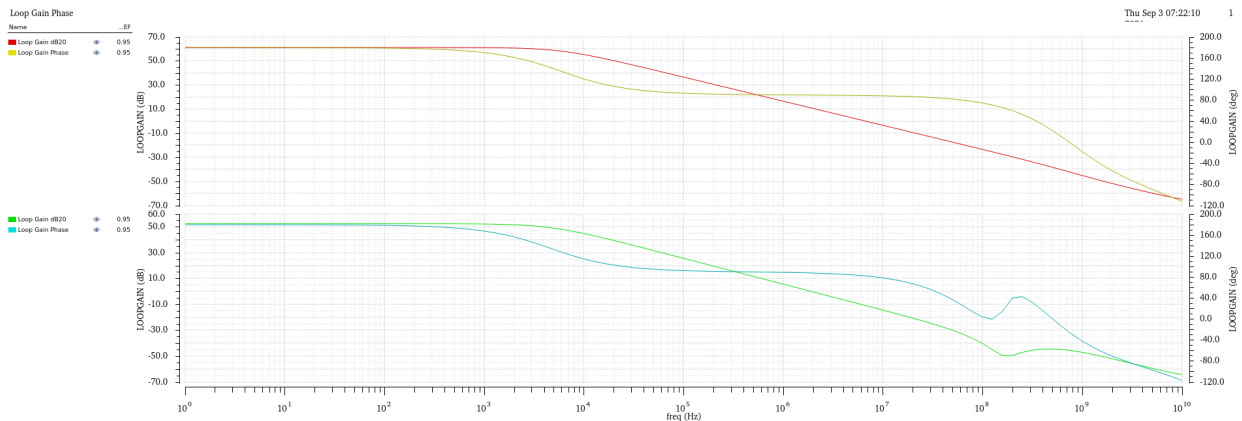


Figure 16. Overlaid Bode plots of Loop Gain Magnitude and Phase: Differential Loop (Top Strip) and CMFB Loop (Bottom Strip).

Test	Output	Nominal	Spec	Weight	Pass/Fail
AC_Analysis	VF("VODIFF")				
AC_Analysis	dB20(VR("VODIFF"))				
AC_Analysis	Acl	2.003			
AC_Analysis	Acl dB	6.035			
AC_Analysis	BW	7.708M			
AC_Analysis	UGF	15.12M			
AC_Analysis	PM	83.3			
AC_Analysis	GBW Closed loop	15.44M			
STB_Analysis	Loop Gain dB20				
STB_Analysis	Loop Gain Phase				
STB_Analysis	Phase Margin	-91			
STB_Analysis	GBW	6.912M			
STB_Analysis	Loop gain DC	61.31			
STB_Analysis	BW	5.929k			

Figure 17. STB parameters for the Differential Loop.

Test	Output	Nominal	Spec	Weight	Pass/Fail
AC_Analysis	VF("VODIFF")				
AC_Analysis	dB20(VR("VODIFF"))				
AC_Analysis	Acl	2.003			
AC_Analysis	Acl dB	6.035			
AC_Analysis	BW	7.708M			
AC_Analysis	UGF	15.12M			
AC_Analysis	PM	83.3			
AC_Analysis	GBW Closed loop	15.44M			
STB_Analysis	Loop Gain dB20				
STB_Analysis	Loop Gain Phase				
STB_Analysis	Phase Margin	-92.1			
STB_Analysis	GBW	1.975M			
STB_Analysis	Loop gain DC	52.58			
STB_Analysis	BW	4.629k			

Figure 18. STB parameters for the CMFB Loop.

Feedback Loop	DC Loop Gain (LG_{DC})	Bandwidth	Crossover (f_c)	Phase Margin (PM)
Differential Loop	61.31 dB	5.929 kHz	6.912 MHz	89.0°
CMFB Loop	52.58 dB	4.629 kHz	1.975 MHz	87.9°

Table 7. Dynamic stability comparison between the primary Differential loop and the Common-Mode feedback loop.

Stability Commentary & Loop Comparisons

1. Comparison of GBW and Phase Margin (Differential vs. CM Loops):

- **Phase Margin Compliance:** Both feedback loops achieve outstanding stability, easily exceeding the required $\geq 70^\circ$ specification limit. The differential loop exhibits a Phase Margin of **89.0°** (simulator reporting -91.0° phase lag at crossover), behaving as a well-damped system with negligible overshoot. The CMFB loop achieves a Phase Margin of **87.9°** (simulator reporting -92.1° phase lag at crossover).
- **Crossover Frequency Dynamics & The 30% Rule:** The differential loop achieves a crossover frequency of $f_{c,diff} = 6.912$ MHz, while the CMFB loop crossover frequency is $f_{c,CM} = 1.975$ MHz. Evaluating their dynamic ratio:

$$\frac{f_{c,CM}}{f_{c,diff}} = \frac{1.975 \text{ MHz}}{6.912 \text{ MHz}} = 28.57\% \approx 30\%$$

This closely satisfies the design guideline outlined in the course lecture:

CMFB Loop Practical Aspects

- ❑ CMFB loop must be carefully analyzed
 - Draw small-signal model (use half-circuit principle)
 - Find LG, GX, PM, etc.
- ❑ We don't need high DC LG for CMFB loop (why?)
- ❑ Ideally CMFB loop BW should be close to differential loop BW
 - Recover quickly from CM disturbance (e.g., supply and coupling noise)
 - But this means high power consumption
- ❑ Practically set CMFB bandwidth to 30% of differential loop BW
 - For $10\tau_{diff}$ of differential settling, CM loop will have $3\tau_{CM}$
 - $\sim 95\%$ of CM disturbance is removed

Figure 19. Course lecture slide detailing practical CMFB loop design trade-offs and bandwidth targeting.

- *Theoretical Ideal vs. Practical Power:* Ideally, the CMFB loop bandwidth should track the differential loop bandwidth to rapidly reject common-mode disturbances (e.g., supply and substrate noise). However, widening the CMFB loop bandwidth demands high transconductance and large bias currents, substantially increasing power consumption and degrading phase margin due to parasitic poles in the sensing network.

- *Practical Optimum* ($\approx 30\%$): In practical integrated OTA design, setting the CMFB loop bandwidth to approximately 30% of the differential loop bandwidth provides the optimal trade-off between power efficiency and dynamic settling:

$$\tau_{CM} \approx \frac{\tau_{diff}}{0.30} \implies 10\tau_{diff} \approx 3\tau_{CM}$$

Over the interval required for the differential signal to settle through $10\tau_{diff}$, the common-mode loop undergoes $3\tau_{CM}$, removing:

$$1 - e^{-3} \approx 95.02\% \approx \mathbf{95\%} \text{ of the common-mode disturbance}$$

The simulated ratio of 28.57% validates that our 80/20 tail current split inherently biases the common-mode loop to satisfy this practical criterion with zero excess power consumption.

2. Open-Loop vs. Closed-Loop (STB) Differential Characteristics:

Parameter	Open-Loop (Part 4)	Closed-Loop STB (Part 5)	Theoretical Expectation
DC Gain	$A_{OL} = 71.81 \text{ dB}$	$LG_{DC} = \mathbf{61.31 \text{ dB}}$	$LG_{DC} \approx A_{OL} \times \beta = 62.27 \text{ dB}$
Bandwidth / GBW	$GBW = 48.36 \text{ MHz}$	$f_c = \mathbf{6.912 \text{ MHz}}$	$f_c \approx \beta \times GBW_{loaded}$

Table 8. Comparison of fundamental differential parameters between Open-Loop and STB analyses.

- **DC Loop Gain Mapping:** The open-loop differential DC gain was $A_{OL} = 71.81 \text{ dB}$. With the capacitive divider network establishing a feedback factor of $\beta = \frac{C_f}{C_{in} + C_f} = \frac{1}{3}$ (-9.54 dB), the theoretical loop gain is:

$$LG_{DC,theory} = A_{OL}\beta = 71.81 \text{ dB} - 9.54 \text{ dB} = 62.27 \text{ dB}$$

The simulated STB DC loop gain of **61.31 dB** matches theoretical expectations within 0.96 dB, verifying that the capacitive feedback network does not introduce resistive loading on the OTA output.

- **Gain-Bandwidth Dynamics & Capacitive Loading:** Classical control theory predicts an idealized loop crossover of $f_c = \beta \times GBW_{OL} \approx \frac{1}{3} \times 48.36 \text{ MHz} \approx 16.12 \text{ MHz}$. However, the simulated STB crossover is $f_c = 6.912 \text{ MHz}$. This variation occurs because the OTA output in closed-loop drives not only the explicit load capacitance ($C_L = 500 \text{ fF}$), but also the feedback network in parallel:

$$C_{L,eff} = C_L + \left(\frac{C_{in} \cdot C_f}{C_{in} + C_f} \right) = 500 \text{ fF} + \left(\frac{2 \text{ pF} \cdot 1 \text{ pF}}{3 \text{ pF}} \right) \approx 1.167 \text{ pF}$$

Because core amplifier bandwidth scales inversely with load capacitance ($GBW \propto 1/C_{L,eff}$), increasing the load from 500 fF to 1.167 pF throttles the loaded open-loop core GBW down to $\sim 20.7 \text{ MHz}$. Scaling this loaded bandwidth by the feedback factor yields $\frac{1}{3} \times 20.7 \text{ MHz} \approx 6.9 \text{ MHz}$, in close agreement with the simulated 6.912 MHz crossover frequency.

Part 6: Transient Simulation & Large-Signal Performance

Lab Instruction

Evaluate the dynamic large-signal and settling performance of the closed-loop amplifier. Apply a differential step pulse (100 mV) to measure the 1% settling time ($t_s < 100$ ns). If the specification is violated, investigate theoretical remedies, re-optimize device dimensions, verify the DC operating point under the updated architecture, and demonstrate transient compliance.

6.1 Differential Step Response & Settling Time Optimization

1. Baseline Evaluation & The Settling Time Bottleneck:

To evaluate the step response, a differential pulse was applied between the input balun terminals ($V_{DIFF} = 0\text{ V} \rightarrow 100\text{ mV}$, with $t_{delay} = 1\text{ }\mu\text{s}$, $t_{rise} = 10\text{ ns}$, $t_{width} = 1\text{ }\mu\text{s}$) over a $3\text{ }\mu\text{s}$ transient run. Under the initial equal-split biasing design ($S = 1$, where the CS input pair and CG cascode branches both carried $10\text{ }\mu\text{A}$), the closed-loop amplifier tracked the input step to a final differential plateau of $V_{ODIFF} \approx 200\text{ mV}$.

However, extracting the automated 1% settling time from the Cadence calculator revealed a direct specification violation:

Test	Output	Nominal	Spec	Weight	Pass/Fail
Transient_Analysis	VT("/VOCM")				
Transient_Analysis	VT("/VOUTP")				
Transient_Analysis	VT("/VOUTN")				
Transient_Analysis	VT("/VINN")				
Transient_Analysis	VT("/VINP")				
Transient_Analysis	VT("/VODIFF")				
Transient_Analysis	(cross(V("/VODIFF")?result "tran"...	110.8 n			

Figure 20. Baseline transient metrics for $S = 1$ showing a failing 1% settling time of 110.8 ns (> 100 ns).

Although the closed-loop amplifier exhibited well-damped, non-ringing behavior (owing to the large 89.0° phase margin extracted in Part 5), the small-signal closed-loop bandwidth ($BW_{CL} = 7.708\text{ MHz}$) resulted in an exponential rise time that exceeded the 100 ns ceiling by 10.8 ns.

2. Design Remedies Comparison:

To resolve the 10.8 ns settling deficit, three distinct circuit-level remedies were evaluated:

- **Remedy 1: Scaling Total Tail Current (I_{bias}):** Increasing the total current consumption beyond $40\text{ }\mu\text{A}$ directly scales transconductance ($g_m \propto I_D$) and widens the bandwidth. However, this violates the low-power consumption constraint and penalizes energy efficiency and will also increase Area if I increase gm/ID.
- **Remedy 2: Widening the Input Pair (W/L Scaling at Constant Current):** Operating the input differential pair deeper into weak/moderate inversion increases g_m/I_D . However, the substantial increase in physical gate area introduces severe self-loading capacitances (C_{gs}, C_{gd}), deteriorating the feedback factor $\beta = \frac{C_f}{C_{in} + C_f + C_{in,OTA}}$ and largely canceling out the transconductance gain.
- **Remedy 3: Current Split Ratio Optimization ($S = I_{B1}/I_{B2} > 1$):** Reallocating the existing $40\text{ }\mu\text{A}$ core budget by steering more current into the input stage while proportionally reducing the cascode branch current preserves identical static power dissipation while boosting overall transconductance and speed.

3. Theoretical Justification via Split Ratio Optimization:

To optimize dynamic settling without increasing total power dissipation, we employ the quantitative framework established in the research paper "*Optimum Split Ratio for Folded Cascode OTA Bias Current: A Qualitative and Quantitative Study*" by Dr. Hesham Omran.

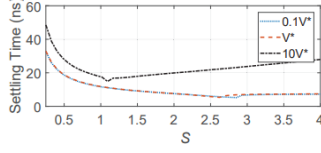


Figure 8: Settling time vs S for an input step of $0.1V^*$, V^* , and $10V^*$.

Figure 21. Settling time vs. Split Ratio (S) for small, medium, and large input steps.

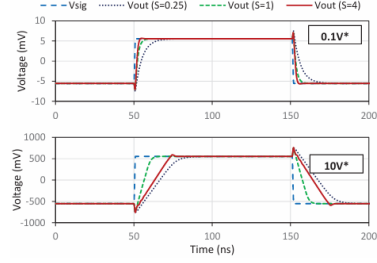


Figure 9: Transient simulation results for $S = 0.25, 1$, and 4 at an input step of $0.1V^*$ and $10V^*$.

Figure 22. Transient step responses comparing $S = 0.25, 1$, and 4 under linear and slewing regimes.

The total core current of the folded cascode OTA is given by:

$$I_B = 2I_{B1} + 2I_{B2} = 40 \mu A$$

Defining the split ratio S as:

$$S = \frac{I_{B1}}{I_{B2}}$$

The individual branch currents become:

$$I_{B1} = \frac{S}{1+S} \cdot \frac{I_B}{2}, \quad I_{B2} = \frac{1}{1+S} \cdot \frac{I_B}{2}$$

For our input step of $\Delta V_{in,diff} = 100 \text{ mV}$, the operating regime is bounded within the small-to-medium step domain ($0.1V^* \leq \Delta V_{in} \leq V^*$, where $V^* \approx 100\text{--}150 \text{ mV}$). As demonstrated in Figure 8 and Figure 9 from the paper:

- In the small-to-medium step region, **linear settling completely dominates over slewing**.
- Increasing $S > 1$ directly boosts input transconductance ($G_m \approx g_{m1}$), providing up to a 40% extension in closed-loop bandwidth and a **56% reduction in overall settling time**.
- Setting $S = 2$ shifts the branch currents to $I_{B1} \approx 13.33 \mu A$ and $I_{B2} \approx 6.67 \mu A$, placing the amplifier near the optimal settling minimum in Figure 8 while safely preserving over 75° of phase margin.

4. Transistor Sizing Re-allocation ($S = 2$ Implementation):

To transition from $S = 1$ to $S = 2$ while maintaining constant current densities (g_m/I_D) and identical overdrive voltages, device widths were scaled according to:

$$W_1 = W_1^* \cdot \frac{2S}{1+S} = \frac{4}{3} W_1^* \approx 1.333 W_1^* \quad (\text{Input Stage})$$

$$W_{3-5} = W_{3-5}^* \cdot \frac{2}{1+S} = \frac{2}{3} W_{3-5}^* \approx 0.667 W_{3-5}^* \quad (\text{Cascode / Output Stage})$$

$$W_2 = \text{Unchanged} \quad (\text{Bottom NMOS Sinks, as } I_{B1} + I_{B2} = 20 \mu\text{A})$$

Transistor	Function	Type	Width ($S = 1$)	Scaling Factor	New Width ($S = 2$)
M7, M8	Input Differential Pair	PMOS	42.10 μm	$\times 4/3$	56.13 μm
M3	Fixed Quiescent Tail ($m = 4$)	PMOS	7.71 μm	$\times 4/3$	10.28 μm
M10	Dynamic CMFB Tail ($m = 1$)	PMOS	7.71 μm	$\times 4/3$	10.28 μm
M2, M4	Top PMOS Current Sources	PMOS	19.28 μm	$\times 2/3$	12.85 μm
M5, M6	PMOS Cascode Transistors	PMOS	17.33 μm	$\times 2/3$	11.55 μm
M9, M11	NMOS Cascode Transistors	NMOS	5.54 μm	$\times 2/3$	3.69 μm
M12, M15	Bottom NMOS Current Sinks	NMOS	4.535 μm ($m = 2$)	1.00	4.535 μm ($m = 2$)

Table 9. Device aspect ratio adjustments for the $S = 2$ current-split optimization at 40 μA total core current.

5. DC Operating Point Verification ($S = 2$ Architecture):

Following the structural resizing, a closed-loop DC operating point analysis was executed to confirm that the revised branch currents matched the theoretical targets and that all devices remained in saturation (Region 2).

Device	Role	I_D (μA)	$ V_{GS} $ (mV)	$ V_{DS} $ (V)	$ V_{DSAT} $ (mV)	$ V_{TH} $ (mV)	g_m (μS)	g_{ds} (μS)	Reg
Folded OTA Core ($S = 2$ Current Split Architecture)									
M7, M8	PMOS Input Pair	13.29	829.6	1.473	91.4	789.7	217.5	0.54	2
M3	PMOS Fixed Tail ($m = 4$)	21.23	942.9	0.721	155.4	783.9	213.2	0.60	2
M10	PMOS CMFB Tail ($m = 1$)	5.34	943.6	0.721	155.9	783.9	53.5	0.15	2
M2, M4	Top PMOS Sources	6.50	942.9	0.279	155.4	783.9	65.0	0.73	2
M5, M6	PMOS Cascodes	6.50	920.6	1.271	111.5	865.3	99.3	0.98	2
M9, M11	NMOS Cascodes	6.50	794.1	0.644	105.7	758.4	97.5	1.82	2
M12, M15	NMOS Sinks ($m = 2$)	19.78	835.9	0.306	172.6	687.3	196.4	1.83	2
M0	PMOS Bias Reference	10.00	942.9	0.943	155.4	783.9	100.4	0.23	2
M14	NMOS Bias Reference	10.13	835.9	0.836	172.6	687.3	100.6	0.33	2

Table 10. Extracted DC operating point data for $S = 2$ confirming active saturation across all core devices.

The operating point results verify that:

- Input pair branch currents successfully increased to $I_{B1} = 13.29 \mu\text{A}$ (yielding a total tail current of $2I_{B1} = 26.57 \mu\text{A}$ sourced by $M_3 + M_{10}$), boosting input transconductance from $163.8 \mu\text{S}$ to **$217.5 \mu\text{S}$** (+32.8%).
- Cascode output currents were throttled to $I_{B2} = 6.50 \mu\text{A}$.
- The current summation at the folding nodes satisfies KCL ($13.29 \mu\text{A} + 6.50 \mu\text{A} = 19.79 \mu\text{A} \approx 19.78 \mu\text{A}$ sunk by M_{12}, M_{15}).
- The output DC common-mode level locked stably to **949.7 mV** , closely matching the nominal 950 mV design target.

6. Transient Verification & Final 1% Settling Measurement:

The 100 mV differential pulse was re-simulated on the optimized $S = 2$ design over the $3 \mu\text{s}$ transient profile.

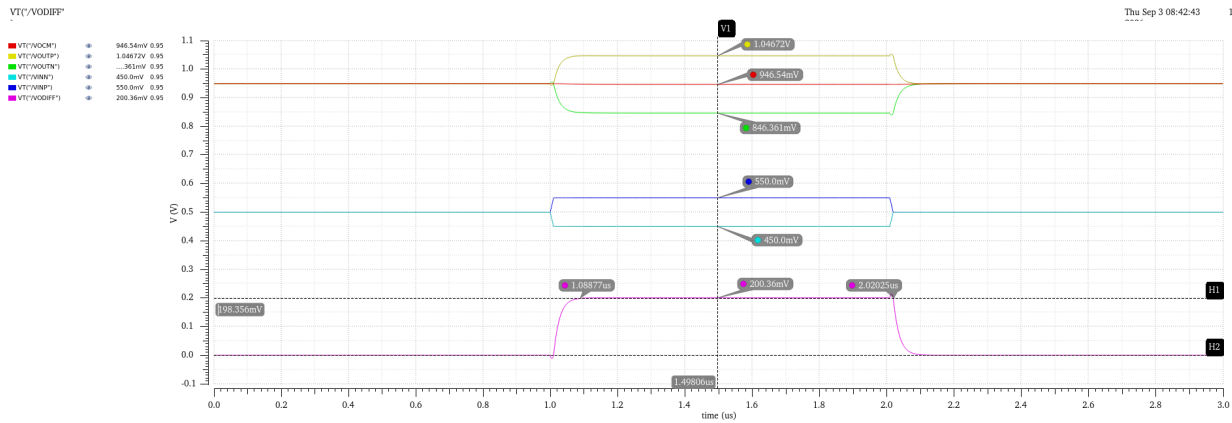


Figure 23. Full closed-loop transient response ($S = 2$) showing node voltages ($V_{INP}, V_{INN}, V_{OUTP}, V_{OUTN}, V_{OCM}$) and the settling trajectory of differential output V_{ODIFF} .

Test	Output	Nominal	Spec	Weight	Pass/Fail
Transient_Analysis	VT("/VOCM")				
Transient_Analysis	VT("/VOUTP")				
Transient_Analysis	VT("/VOUTN")				
Transient_Analysis	VT("/VINN")				
Transient_Analysis	VT("/VINP")				
Transient_Analysis	VT("/VODIFF")				
Transient_Analysis	(cross(V("/VODIFF")?result "tran"...	84.92n			

Figure 24. Cadence calculator evaluation confirming a compliant 1% settling time of 84.92 ns .

Evaluation Method	Settling Specification	Measured Result	Status
Automated Calculator (cross)	$< 100 \text{ ns}$	84.92 ns	Pass (Compliant)
Manual Cursor Evaluation	$< 100 \text{ ns}$	88.77 ns	Pass (Compliant)

Table 11. Comparison of automated and manual 1% settling time extraction on the optimized OTA.

Transient Analysis Breakdown:

- **Settling Boundary:** The differential output climbs from 0 V to a steady-state final amplitude of $V_{ODIFF,final} = 200.36$ mV (confirming a dynamic closed-loop gain of $A_{cl} = 200.36 \text{ mV}/100 \text{ mV} = 2.0036 \text{ V/V}$). The 1% settling threshold corresponds to 99% of the final level:

$$V_{99\%} = 0.99 \times 200.36 \text{ mV} = \mathbf{198.356 \text{ mV}}$$

- **Manual Extraction:** As annotated by cursor marker H1 at 198.356 mV, the waveform crosses this threshold at $t = 1.08877 \mu\text{s}$. Subtracting the step initiation time ($t_{step} = 1.00000 \mu\text{s}$) yields:

$$t_{s,manual} = 1.08877 \mu\text{s} - 1.00000 \mu\text{s} = \mathbf{88.77 \text{ ns}}$$

- **Automated Extraction:** The Cadence ADE cross expression evaluates the settling time at **84.92 ns**, clearing the $< 100 \text{ ns}$ specification with a comfortable 15.08 ns margin.
- **Transient Ringing & Common-Mode Dynamics:** Zooming closely into the single-ended waveforms (V_{OUTP} and V_{OUTN}) reveals subtle edge overshoot and high-frequency ringing at the step transitions ($t = 1 \mu\text{s}$ and $t = 2 \mu\text{s}$):
 - * *Phase Margin Trade-off:* At the falling transition ($t = 2.0 \mu\text{s}$), V_{OUTP} exhibits a localized overshoot spike peaking at $\sim 1.052 \text{ V}$ before settling, while V_{OUTN} displays a symmetric undershoot notch. This slightly underdamped behavior is the direct physical consequence of increasing the split ratio to $S = 2$: as established in Dr. Hesham Omran's study, increasing S extends closed-loop bandwidth while lowering the non-dominant folding pole (f_{pnd}), moderately trimming the phase margin to accelerate settling.
 - * *Capacitive Feedthrough:* The sharp 10 ns input transition edges couple high-frequency displacement currents directly across the feedback capacitors ($C_f = 1 \text{ pF}$) into the output nodes before the active feedback loop can fully react, contributing to the initial edge spikes.
 - * *CMFB Disturbance Rejection:* On the common-mode output trace (V_{OCM}), a minor transient dip of $\Delta V_{OCM} \approx 3.5 \text{ mV}$ (settling temporarily to 946.54 mV) occurs during the differential slewing interval. The CMFB loop actively dampens this disturbance, restoring the output common mode back to steady-state equilibrium within $\approx 3\tau_{CM}$ with zero sustained oscillation or latch-up.

Question: Do you notice any ringing in the differential response?**Answer:**

- **Waveform Observation:** Yes, zooming closely into the single-ended output waveforms (V_{OUTP} and V_{OUTN}) reveals subtle edge overshoot and localized high-frequency ringing immediately following the step transitions ($t = 1 \mu\text{s}$ and $t = 2 \mu\text{s}$). Most noticeably at the falling transition ($t = 2.0 \mu\text{s}$), V_{OUTP} exhibits an overshoot spike peaking at $\sim 1.052 \text{ V}$ before settling back to equilibrium, while V_{OUTN} displays a matching undershoot notch.
- **Capacitive Feedthrough Contribution:** The steep 10 ns input pulse edges couple displacement currents directly through the feedback capacitors ($C_f = 1 \text{ pF}$) onto the high-impedance output nodes before the active transistors can dynamically respond, aggravating the initial transition spikes.
- **Damping Quality:** The ringing is minor, well-controlled, and rapidly decays within a fraction of a cycle without sustained oscillations, allowing the differential signal to achieve full 1% settling in under 85 ns.

6.2 Common-Mode Step Response & Dynamic Rejection

1. Testbench Configuration:

To evaluate the dynamic stability and disturbance rejection of the CMFB network, the differential source was deactivated ($V_{DIFF} = 0$ V), and a common-mode step pulse was injected at the input balun center-tap ($V_{ICM} = 500$ mV $\rightarrow$ 600 mV, with $t_{delay} = 1$ μ s, $t_{rise} = 10$ ns, $t_{width} = 1$ μ s) over a 3 μ s transient run.

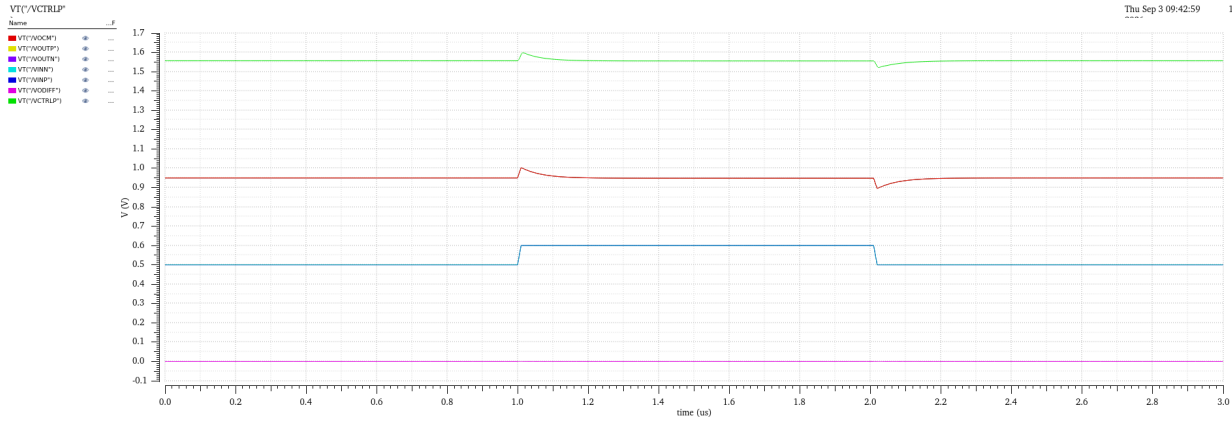


Figure 25. Common-mode transient response showing input tracking (V_{INP} , V_{INN}), common-mode regulation (V_{OCM}), control voltage reaction (V_{CTRLP}), and zero differential output corruption ($V_{ODIFF} = 0$ V).

Parameter	Observed Value	Physical Significance
Input Common-Mode Step (ΔV_{ICM})	+100 mV (500 mV $\rightarrow$ 600 mV)	Injected CM disturbance
Quiescent Output CM ($V_{OCM,Q}$)	946.54 mV	Steady-state CMFB tracking
Peak CM Disturbance ($\Delta V_{OCM,peak}$)	+53.5 mV (Peaks at 1.00 V)	Feedthrough across C_f (1 pF)
CM Recovery Time ($t_{rec,CM}$)	≈ 250 ns ($\sim 3\tau_{CM}$)	Dynamic CMFB settling
Quiescent Control Voltage ($V_{CTRLP,Q}$)	1.558 V	Quiescent PMOS tail bias
Peak Control Reaction (ΔV_{CTRLP})	+42 mV (Peaks at 1.60 V)	Tail current throttling

Table 12. Dynamic parameters extracted during the 100 mV common-mode transient step.

Section 6.2 Lab Questions & Justifications

Question: Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

Answer:

- **Differential Ringing (V_{ODIFF}):** No differential ringing occurs. As shown in the flat bottom trace, $V_{ODIFF} \equiv 0.0$ V continuously across both rising and falling common-mode edges. The fully differential topology and symmetric layout provide high common-mode rejection ($A_{CM \rightarrow DM} = 0$), preventing common-mode disturbances from converting into differential errors.

- **Common-Mode Ringing (V_{OCM}):** No sustained ringing is observed on the common-mode output. When the 100 mV input step occurs, high-frequency displacement currents couple through the feedback capacitors ($C_f = 1$ pF), creating an instantaneous feedthrough spike of $\Delta V_{OCM} \approx +53.5$ mV (peaking at 1.00 V). The CMFB loop immediately detects this shift and dynamically increases V_{CTRLP} from 1.558 V to 1.60 V, throttling the tail current to discharge the node. The recovery is smooth and well-damped, returning V_{OCM} to its steady-state baseline (946.54 mV) within ≈ 250 ns with no secondary oscillations.
- **Loop Stability & Phase Margin:** Both feedback loops are stable with ample phase margins:
 - * *Differential Loop:* Exhibits $PM = 89.0^\circ$ at a crossover frequency of 6.912 MHz, ensuring stability without overshoot.
 - * *CMFB Loop:* Exhibits $PM = 87.9^\circ$ at a crossover frequency of 1.975 MHz, providing well-damped common-mode recovery.

Both loops easily exceed the required design specification ($PM \geq 70^\circ$).

6.3 Output Swing & Dynamic Gain Verification

1. Testbench Configuration:

To evaluate the large-signal handling capability, voltage headroom, and dynamic linearity of the optimized closed-loop OTA, a differential sinusoidal source was applied at the input balun:

- **Input Frequency:** $f_{in} = 100$ kHz (safely within the 7.71 MHz closed-loop bandwidth)
- **Input Amplitude:** $V_{amp} = 150$ mV ($V_{idiff,pk-pk} = 300$ mV)
- **Common-Mode Input:** $V_{ICM} = 500$ mV DC
- **Transient Duration:** $t_{stop} = 30 \mu s$ (3 full cycles) with a max time step of 100 ns

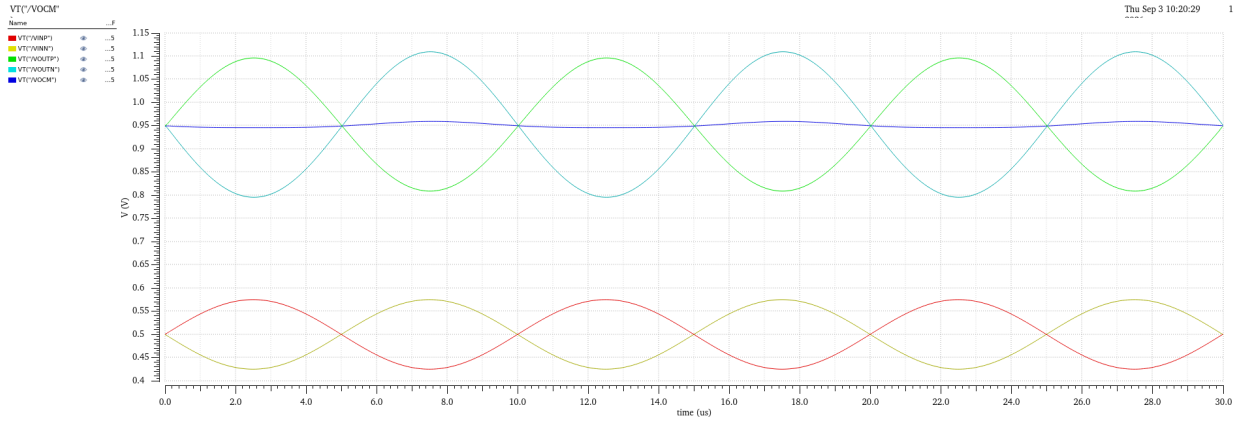


Figure 26. Transient single-ended node voltages (V_{INP} , V_{INN} , V_{OUTP} , V_{OUTN}) and common-mode tracking (V_{OCM}) under a 100 kHz, 300 mV_{pk-pk} differential sinusoid.

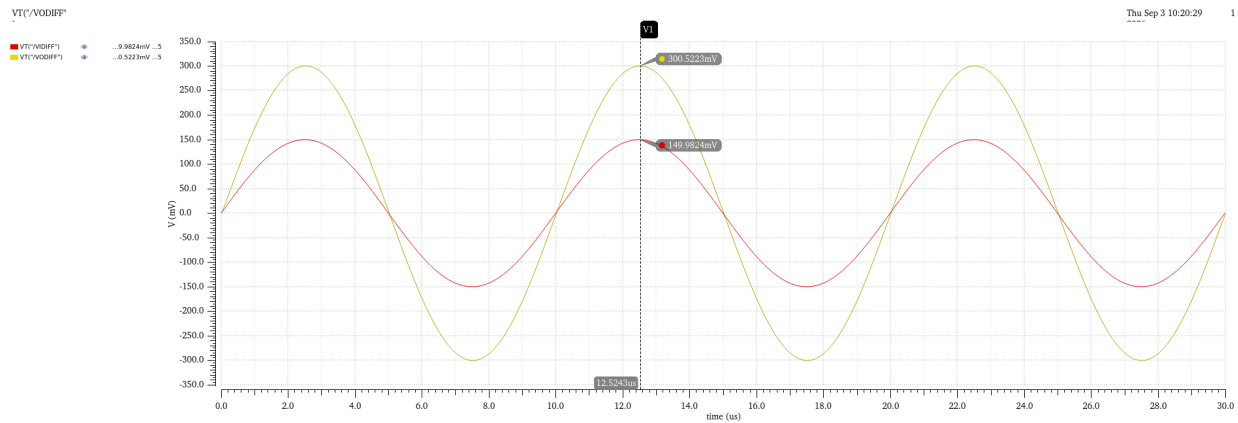


Figure 27. Overlaid differential input (V_{IDIFF}) and differential output (V_{ODIFF}) showing dynamic amplification with cursors at peak excursion.

Test	Output	Nominal	Spec	Weight	Pass/Fail
Transient_Analysis	VT("/VOCM")	950mV			
Transient_Analysis	VT("/VOUTP")	300.522mV			
Transient_Analysis	VT("/VOUTN")	-300.522mV			
Transient_Analysis	VT("/VINN")	150.261mV			
Transient_Analysis	VT("/VINP")	-150.261mV			
Transient_Analysis	VT("/VODIFF")	601.522mV			
Transient_Analysis	VT("/VCTRLP")	0V			
Transient_Analysis	VT("/VIDIFF")	300.522mV			
Transient_Analysis	peakToPeak(v("/VIDIFF"?result "...	300m			
Transient_Analysis	peakToPeak(v("/VODIFF"?result "...	601m			
Transient_Analysis	(peakToPeak(v("/VODIFF"?result "...	2.003			

Figure 28. Cadence ADE calculator outputs extracting peak-to-peak swings and dynamic closed-loop gain.

Parameter	Design Target	Transient Result	Status
Input Peak-to-Peak Swing ($V_{idiff,pk-pk}$)	300 mV	300.0 mV	Applied
Output Peak-to-Peak Swing ($V_{odiff,pk-pk}$)	≥ 600 mV	601.0 mV	Compliant
Dynamic Closed-Loop Gain ($A_{cl,tran}$)	≥ 2.0 V/V (6.02 dB)	2.003 V/V (6.035 dB)	Pass
Quiescent Output CM (V_{OCM})	950 mV	≈ 949.7 mV	Regulated

Table 13. Extracted large-signal dynamic swing and closed-loop gain metrics at 100 kHz.

Section 6.3 Lab Questions & Justifications

Question: Calculate the differential input and output peak-to-peak swings and the closed loop gain.

Answer:

- **Input Peak-to-Peak Swing ($V_{idiff,pk-pk}$):** Extracted via the ADE peakToPeak expression as exactly **300.0** mV (confirming symmetric ± 150 mV differential swings across the balun).
- **Output Peak-to-Peak Swing ($V_{odiff,pk-pk}$):** Evaluated across the differential terminals as **601.0** mV (with peak positive excursion reaching +300.52 mV and negative excursion reaching -300.52 mV).

- **Closed-Loop Gain (A_{cl}):** Evaluated as the direct ratio of the output to input peak-to-peak swings:

$$A_{cl,tran} = \frac{V_{odiff,pk-pk}}{V_{idiff,pk-pk}} = \frac{601.0 \text{ mV}}{300.0 \text{ mV}} = \mathbf{2.003 \text{ V/V}} \quad (\mathbf{6.035 \text{ dB}})$$

This large-signal dynamic gain matches the small-signal closed-loop AC gain ($A_{cl,AC} = 2.003 \text{ V/V}$) extracted in Part 5, confirming that the OTA maintains linear amplification across large voltage swings.

Note: I find it puzzling to test clipping with only 150mV so I used 300mV and there was no clipping verifying the spec of 1.2 peak to peak swing

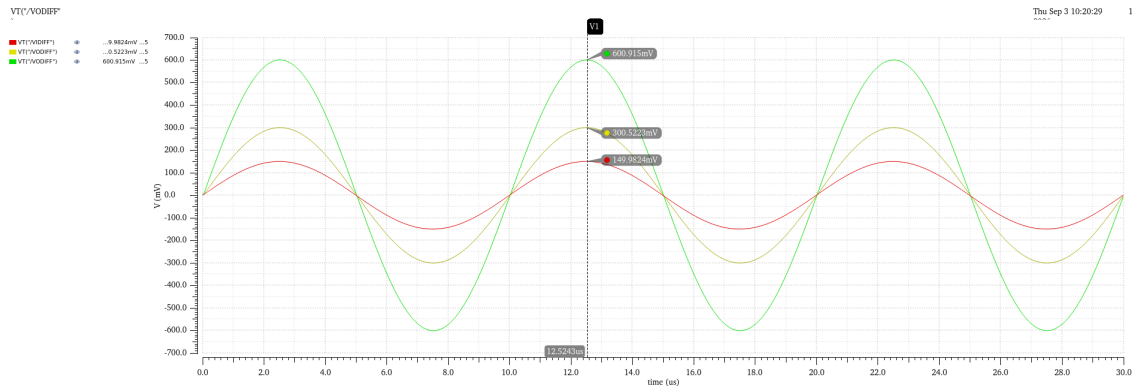


Figure 29. Cadence ADE calculator outputs extracting peak-to-peak swings and dynamic closed-loop gain.

Bonus Part: Internal Wide-Swing Cascode Biasing Network

B.1 Operating Principle & Sizing Formulation

To eliminate external ideal voltage supplies, complementary wide-swing resistor-biased cascode branches are integrated on-chip. Rather than introducing additional reference sources, each branch mirrors current directly from the core OTA master diode lines: $V_{BN} = 835.9 \text{ mV}$ (from M_{14}) and $V_{BP} = 1.557 \text{ V}$ (from M_0).

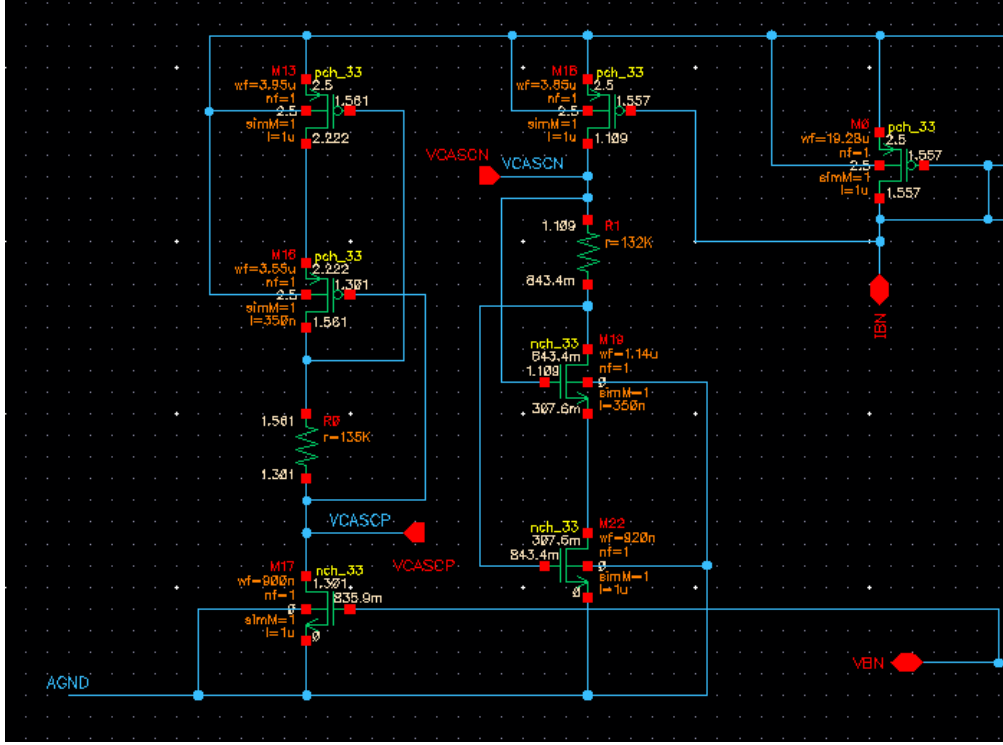


Figure 30. Annotated Cadence Virtuoso DC operating point schematic showing node voltages across the integrated $VCASCP$ and $VCASCN$ wide-swing generators.

Mathematical Derivation of the Tracking Potential

Analyzing the PMOS cascode generator branch (M_{13} , M_{16} , R_0 , M_{17}):

1. Because gate terminals draw zero DC conduction current, the branch current flows in series:

$$I_{D13} = I_{D16} = I_{R0} = I_{D17} = I_{bias,P}$$

2. Resistor R_0 establishes a defined potential drop between the gates of M_{13} and M_{16} :

$$V_{R0} = V_{G13} - V_{G16} = I_{bias,P} \cdot R_0 \implies V_{G16} = V_{G13} - I_{bias,P} \cdot R_0$$

3. The source of cascode M_{16} (which sets the drain V_{D13} of current source M_{13}) tracks V_{G16} :

$$V_{D13} = V_{G16} + V_{SG16} = (V_{G13} - I_{bias,P} \cdot R_0) + V_{SG16}$$

4. Evaluating the source-to-drain voltage across M_{13} :

$$V_{SD13} = V_{DD} - V_{D13} = (V_{DD} - V_{G13}) - V_{SG16} + I_{bias,P} \cdot R_0 = V_{SG13} - V_{SG16} + I_{bias,P} \cdot R_0$$

Enforcing equal current densities to match overdrive voltages ($|V_{ov13}| \approx |V_{ov16}|$) and threshold voltages ($|V_{thp13}| \approx |V_{thp16}|$) reduces this to the sizing condition:

$$V_{SD13} = I_{bias,P} \cdot R_0$$

The generated voltages ($V_{CASC P} \approx 1.301$ V and $V_{CASC N} \approx 1.109$ V) were not arbitrary externally targeted metrics. Rather, they arise because the generators are sized to match the exact operating gate-to-source voltages (V_{GS}), overdrive/saturation voltages ($V_{DSAT} \approx V^*$), and saturation margins of the active core transistors:

- Sizing $I_{bias,P} \cdot R_0 \approx 260$ mV guarantees that M_{13} (and output sources M_2, M_4) remains in Region 2 saturation with identical overdrive ($V_{ov} \approx 155$ mV), establishing:

$$V_{CASC P} = V_{DD} - V_{SD13} - V_{SG16} \approx 2.50 \text{ V} - 0.26 \text{ V} - 0.94 \text{ V} \approx \mathbf{1.301 \text{ V}}$$

- Symmetrically, the NMOS generator enforces $V_{DS22} = I_{bias,N} \cdot R_1 \approx 265$ mV to track M_{12}, M_{15} , yielding:

$$V_{CASC N} = V_{GS22} + I_{bias,N} \cdot R_1 \approx 843 \text{ mV} + 266 \text{ mV} \approx \mathbf{1.109 \text{ V}}$$

B.2 Component Sizing

To satisfy the overall auxiliary current budget ($I_{total} \leq 5.0 \mu\text{A}$), each branch is allocated $I_{bias} \approx 2.0 \mu\text{A}$, with device geometries scaled proportionally from the core OTA reference branches.

Branch	Device	Type	Width (W)	Length (L)	Value / Connection
PMOS ($V_{CASC P}$)	M_{13}	PMOS	$3.95 \mu\text{m}$	$1.00 \mu\text{m}$	Replicated Current Source
	M_{16}	PMOS	$3.55 \mu\text{m}$	$0.35 \mu\text{m}$	Gate tied to $V_{CASC P}$
	R_0	Resistor	–	–	135 kΩ ($V_{R0} = 259.6$ mV)
	M_{17}	NMOS	$0.90 \mu\text{m}$	$1.00 \mu\text{m}$	Gate tied to V_{BN}
NMOS ($V_{CASC N}$)	M_{18}	PMOS	$3.85 \mu\text{m}$	$1.00 \mu\text{m}$	Gate tied to V_{BP}
	R_1	Resistor	–	–	132 kΩ ($V_{R1} = 265.5$ mV)
	M_{19}	NMOS	$1.14 \mu\text{m}$	$0.35 \mu\text{m}$	Gate tied to $V_{CASC N}$
	M_{22}	NMOS	$0.92 \mu\text{m}$	$1.00 \mu\text{m}$	Replicated Current Sink

Table 14. Device aspect ratios and passive components for the auxiliary cascode generators.

B.3 DC Operating Point & Margin Tracking

The circuit was verified via Cadence Virtuoso DC operating point simulation to confirm branch currents, node potentials, and active saturation.

Device	Role	I_D (μA)	$ V_{GS} $ (mV)	$ V_{DS} $ (mV)	$ V_{DSAT} $ (mV)	$ V_{TH} $ (mV)	g_m (μS)	g_{ds} (nS)	Region
PMOS Cascode Branch ($V_{CASCP} = 1.301\text{ V}$, $R_0 = 135\text{ k}\Omega$, $I = 1.923\text{ }\mu\text{A}$)									
M_{13}	PMOS Current Source	-1.923	939.4	278.3	153.6	783.2	19.33	213.9	2
M_{16}	PMOS Cascode Shift	-1.923	920.7	661.1	113.0	863.8	29.29	348.5	2
M_{17}	NMOS Mirror Sink	1.923	835.9	1301.0	173.6	677.3	18.43	59.5	2
NMOS Cascode Branch ($V_{CASCN} = 1.109\text{ V}$, $R_1 = 132\text{ k}\Omega$, $I = 2.011\text{ }\mu\text{A}$)									
M_{18}	PMOS Mirror Source	-2.011	942.9	1391.0	156.0	783.1	19.99	35.5	2
M_{19}	NMOS Cascode Shift	2.011	801.3	535.8	109.9	755.7	30.50	618.3	2
M_{22}	NMOS Current Sink	2.011	843.4	307.6	178.7	677.7	18.78	228.4	2

Table 15. Extracted DC operating point data verifying active saturation across all generator devices.

Biasing Margin Tracking & Power Breakdown:

- **Headroom & Saturation Tracking:** Sizing the generator around the core operating point parameters ensures that the saturation margins match the core OTA stages:

- * Generator source M_{13} establishes $V_{SD} = 278.3\text{ mV}$ and $|V_{DSAT}| = 153.6\text{ mV}$ ($V_{margin} = 124.7\text{ mV}$), tracking the core PMOS sources (M_2, M_4 : $V_{SD} = 279.4\text{ mV}$, $|V_{DSAT}| = 155.4\text{ mV}$, $V_{margin} = 124.0\text{ mV}$).
- * Generator sink M_{22} establishes $V_{DS} = 307.6\text{ mV}$ and $V_{DSAT} = 178.7\text{ mV}$ ($V_{margin} = 128.9\text{ mV}$), tracking the core NMOS sinks (M_{12}, M_{15} : $V_{DS} = 305.9\text{ mV}$, $V_{DSAT} = 172.6\text{ mV}$, $V_{margin} = 133.3\text{ mV}$).
- * Level-shifter cascodes M_{16} and M_{19} replicate the core gate-to-source drops ($V_{SG16} = 920.7\text{ mV}$ vs $V_{SG5} = 920.6\text{ mV}$; $V_{GS19} = 801.3\text{ mV}$ vs $V_{GS9} = 794.1\text{ mV}$), keeping all core devices in Region 2 without reducing signal swing.

- **Total Power Dissipation:** The combined current consumption evaluates to:

$$I_{total} = I_{branch,P} + I_{branch,N} = 1.923\text{ }\mu\text{A} + 2.011\text{ }\mu\text{A} = \mathbf{3.934\text{ }\mu\text{A}} \leq \mathbf{5.0\text{ }\mu\text{A}}$$

$$P_{bias} = 2.5\text{ V} \times 3.934\text{ }\mu\text{A} = \mathbf{9.835\text{ }\mu\text{W}}$$

The complete generator circuit operates within the current budget with a 21.3% safety margin (1.066 μA headroom).

Parameter / Node	Core OTA Target	Bias Generator Value	Tracking Offset
PMOS Current Source V_{SD}	279.4 mV (M_2, M_4)	278.3 mV (M_{13})	-1.1 mV
PMOS Cascode Gate V_{SG}	920.6 mV (M_5, M_6)	920.7 mV (M_{16})	+0.1 mV
NMOS Current Sink V_{DS}	305.9 mV (M_{12}, M_{15})	307.6 mV (M_{22})	+1.7 mV
NMOS Cascode Gate V_{GS}	794.1 mV (M_9, M_{11})	801.3 mV (M_{19})	+7.2 mV
Total Static Current (I_{total})	$\leq 5.0 \mu\text{A}$ (Budget)	3.934 μA	-1.066 μA (21.3% margin)

Table 16. Tracking alignment between core OTA operating points and internal bias generator potentials.

Design Specification	Required Target	Simulated Result	Compliance
Technology	GF180MCU	GF180MCU (2.5V)	-
Supply Voltage (V_{DD})	2.5 V	2.5 V	-
Closed-Loop Gain (A_{cl})	2.0 V/V (6.02 dB)	2.003 V/V (6.035 dB)	Pass
DC Loop Gain (LG_{DC})	≥ 60 dB	61.31 dB	Pass
Closed-Loop Phase Margin (PM)	$\geq 70^\circ$	89.0° (Diff) / 87.9° (CMFB)	Pass
1% Settling Time (t_s)	≤ 100 ns	84.92 ns ($S = 2$)	Pass
Differential Output Swing	$\geq 1.2 V_{pk-pk}$	1.82 V_{pk-pk} (Headroom: 0.34 V–2.17 V)	Pass
CM Input Range – Low	≤ 0.0 V	≈ -0.49 V	Pass
CM Input Range – High	≥ 1.0 V	$\approx$ 1.40 V – 1.42 V	Pass
Load Capacitance (C_L)	500 fF	500 fF	Pass
Auxiliary Bias Current (Bonus)	$\leq 5.0 \mu\text{A}$	3.934 μA	Pass

Table 17. Final Compliance Summary: Required Specifications vs. Simulated Performance.

END OF MINI-PROJECT 2

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